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Beyond ratings: deconstructing ESG disclosure through AI-driven action-level knowledge graph analysis

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ABSTRACT

Third-party environmental, social and governance (ESG) ratings dominate corporate sustainability evaluation despite low inter-rater reliability (average correlation 0.61) and methodological opacity that obscures underlying evidence and weighting assumptions. This study uses large language models (LLMs) to extract granular action records from corporate disclosures and constructs knowledge graphs encoding relationships between companies, topics, stakeholders and activities. From 2024 sustainability reports of 141 Japanese listed firms, analysis extracted 77,734 action records comprising eight elements (company, topic, actor, action, target, modality, evidence and confidence), with 97.7% exceeding 0.9 confidence.

Analysis reveals patterns aggregate ratings obscure. Japanese firms emphasize social disclosure (55.4%) over environmental (26.9%), prioritize ongoing implementation (65.5%) over future commitments (19.1%), and concentrate stakeholder attention on employees rather than external groups. Network analysis identifies topic co-occurrence structures and three distinct disclosure archetypes aligned with industry materiality. This framework complements evaluative ratings by enabling source-traceable evidence review, user-customizable materiality weighting and systematic disclosure pattern analysis – capabilities proprietary rating methodologies do not provide. Methodologically, this study contributes scalable infrastructure for action-level ESG analysis. Theoretically, this approach operationalizes stakeholder salience and signaling credibility through empirical extraction of actor–target relationships and temporal commitment patterns.

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1. Introduction

Environmental, social and governance (ESG) disclosure links corporate sustainability practices with stakeholder decisions (Dhaliwal et al., 2011; Ioannou & Serafeim, 2017). High-quality disclosure reduces information asymmetry (Dhaliwal et al., 2011), correlates with superior long-term financial performance (Eccles et al., 2014; Whelan et al., 2021), and shapes strategic decision-making (Eccles et al., 2014). Yet the dominant evaluation paradigm – third-party rating agencies – faces mounting credibility concerns.

Christensen et al. (2022) identify a 'disclosure paradox': increased disclosure volume paradoxically amplifies rating divergence through greater interpretive flexibility. Different agencies assign vastly different ESG scores to identical companies. Berg et al. (2022) document average inter-rater correlations of only 0.61 (range: 0.38–0.71) across six major providers – far below the 0.92–0.99 consensus among credit raters. Their decomposition attributes 56% of divergence to measurement differences, 38% to scope differences and 6% to weighting.

This divergence need not signal system failure. Different stakeholders (investors, regulators, NGOs) legitimately prioritize different ESG dimensions according to their decision contexts (Amel-Zadeh & Serafeim, 2018; Cao et al., 2023). The challenge lies instead in aggregation opacity: when sustainability practices compress into scalar scores, underlying evidence and weighting assumptions become

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inaccessible. Institutional investors increasingly demand granular ESG information (Amel-Zadeh & Serafeim, 2018; Cao et al., 2023), yet proprietary methodologies prevent customization to portfolio-specific materiality. This opacity motivates the complementary approach developed here.

1.1. Methodological positioning

This approach diverges from traditional ESG performance evaluation. Where conventional ratings ask 'How well does this company perform on ESG dimensions'?, the knowledge graph developed here poses 'What specific actions does this company disclose, and can these claims be traced to supporting evidence'? The distinction centers on disclosure transparency rather than performance judgment.

ESG ratings compress multidimensional sustainability practices into scalar scores through necessarily subjective metric weighting. This study maintains granular action-level detail with full source traceability. Stakeholders can trace specific claims to supporting documents, weight topics according to their own materiality criteria, and distinguish commitment-heavy disclosure from demonstrated implementation. This transparency enables customization impossible with fixed-weight aggregate scores.

The framework does not evaluate whether disclosed actions constitute adequate ESG performance – such judgments require external benchmarks (regulatory standards, science-based targets, industry norms) that vary by context and stakeholder. Rather, it provides infrastructure for evidence-based assessment that complements existing rating systems by exposing the action-level detail they aggregate away.

Large language models (LLMs) offer capabilities unavailable to traditional text analysis methods (Hogan et al., 2022; Ji et al., 2022). Where keyword matching or supervised learning requires extensive labeled data, LLMs extract semantically rich relationships through in-context learning and instruction following. For ESG analysis, this means identifying not merely which topics companies disclose, but who performs actions, what activities occurred, who benefits and whether firms report completed implementation or future commitments. These dimensions – essential for assessing disclosure credibility – remain largely absent from conventional rating systems.

This study examines: How can LLMs extract and analyze action-level ESG disclosures to reveal patterns in corporate sustainability reporting? Specifically:

RQ1: What patterns characterize action-level ESG disclosure – dimensional emphasis (ESG balance), stakeholder structures (actors and targets) and temporal framing (completed actions *versus* future commitments)?
RQ2: How do ESG topics co-occur and integrate within corporate disclosure, and what network structures characterize these patterns?
RQ3: Do firms exhibit distinct disclosure archetypes reflecting industry-specific materiality and strategic priorities?

Japan provides an ideal empirical setting. Its regulatory environment transformed substantially from 2015 to 2021 through Corporate Governance Code reforms (Miyajima & Saito, 2021), the Government Pension Investment Fund's endorsement of Responsible Investment Principles and national carbon neutrality commitments (Government of Japan, 2021) – changes that created de facto mandatory ESG reporting expectations. Japan also faces sustainability pressures common to advanced economies: aging demographics, constrained labor markets and natural disaster vulnerability (Kato & Kodama, 2018). Methodologically, Japanese corporations produce comprehensive sustainability reports in English or bilingually, providing high-quality multilingual data suitable for computational extraction.

This study develops an LLM-driven framework that extracts structured action records from sustainability narratives and constructs knowledge graphs encoding relationships among companies, topics, stakeholders and activities. Rather than examining aggregate scores, the approach investigates disclosed actions, participating stakeholders and whether commitments reflect completed implementation or future plans. This yields transparent, reproducible analysis unavailable through proprietary rating methodologies.

The resulting dataset comprises 77,734 action records from 141 Japanese listed firms – the most extensive action-level ESG corpus for Japan. Analysis reveals patterns aggregate metrics obscure: social

disclosure dominance, action-oriented (rather than commitment-heavy) rhetoric and employee-centric stakeholder structures.

This study makes dual contributions. *Methodologically*, it provides scalable infrastructure for action-level ESG analysis complementing existing rating systems. *Theoretically*, it operationalizes constructs difficult to measure empirically. Extracting actor–target relationships operationalizes stakeholder salience theory (Freeman, 1984; Mitchell et al., 1997), revealing which groups receive disclosed attention. Distinguishing completed actions from future commitments through modality analysis extends signaling theory (Connelly et al., 2011), enabling credibility assessment beyond thematic coverage.

Prior ESG text mining approaches – topic modeling for thematic coverage, BERT-based classification for sentence-level categorization – lack the granularity this knowledge graph provides. The framework enables three analytical capabilities unattainable with coarser methods: stakeholder relationship mapping, credibility assessment *via* modality analysis (distinguishing stated plans from completed actions, Section 3.4) and topic integration network analysis (Section 2.4). Practically, investors, regulators and corporations can perform customizable queries, action-level benchmarking and systematic green-washing detection.

In the following sections: Section 2 reviews literature on ESG disclosure quality, rating divergence and AI-driven text analysis. Section 3 details the data, LLM-based extraction methodology and knowledge graph construction. Section 4 presents descriptive and network analysis results. Section 5 interprets findings theoretically, discusses methodological implications and practical applications, and addresses limitations. Section 6 concludes by synthesizing contributions and outlining strategic directions for extending this framework to longitudinal tracking, multimodal data integration and cross-country comparative analysis.

2. Literature review

2.1. ESG disclosure and the rating divergence problem

Early ESG disclosure research examined binary disclosure decisions (Clarkson et al., 2008) before shifting toward quality assessment and content analysis (Fifka, 2013; Gond et al., 2011). Dhaliwal et al. (2011) demonstrate that initiating CSR reporting associates with reduced equity capital cost, particularly for firms with limited analyst coverage, confirming disclosure's role in mitigating information asymmetry. Whelan et al. (2021) meta-analyzed over 1000 studies, revealing 58% report positive ESG-financial performance associations while only 8% indicate negative relationships, supporting sustainability's financial materiality. Khan et al. (2016) establish that financially material ESG issues associate with abnormal stock returns, influencing industry-specific standards like the Sustainability Accounting Standards Board (SASB). Recent scholarship emphasizes disclosure harmonization across frameworks to reduce interpretive divergence (Grewal & Serafeim, 2020; Lin et al., 2024).

Disclosure quality varies significantly, and its interpretation remains contested. Hummel and Schlick (2016) examine 1952 European firms, revealing entities with poor ESG performance provide more extensive disclosures, suggesting impression management rather than genuine transparency. Christensen et al. (2022) demonstrate increased disclosure intensifies rating divergence as agencies subjectively interpret identical qualitative statements. Luo and Tang (2023) document that GRI standard adoption reduces carbon emissions primarily in jurisdictions with strong enforcement, indicating disclosure effectiveness depends critically on institutional context.

Berg et al. (2022) decompose rating divergence, revealing measurement variance accounts for 56% of total dispersion. Different agencies apply divergent indicators for identical ESG dimensions – 'employee diversity' may be assessed through board gender ratios, workforce demographics, or policy presence, generating inconsistent evaluations. Scope divergence (38%) stems from agencies selecting different issue sets: some prioritize biodiversity and water stewardship, others emphasize cybersecurity and human capital. This 'aggregate confusion' indicates ratings reflect rater methodological preferences as much as firm performance.

Hawn and Ioannou (2016) advocate 'detailed analysis' and 'process tracing' to understand specific corporate actions, their rationales, and interrelationships. LLM-powered knowledge graphs address this methodological gap through large-scale action-level extraction.

2.2. Japan's ESG disclosure context

Japan's ESG disclosure evolved from voluntary to quasi-mandatory frameworks. The 2015 Corporate Governance Code mandated corporations integrate stakeholder interests and ESG considerations into long-term value strategies (Miyajima & Saito, 2021). Government Pension Investment Fund adoption of Responsible Investment Principles in 2017, managing $1.6 trillion, integrated ESG into mainstream finance. Yoshikawa and Rasheed (2009) demonstrate Japanese institutional investors exhibit passive compliance patterns, prioritizing governance and long-term value creation, reflecting stakeholder capitalism traditions.

Japan leads TCFD adoption with over 800 corporate endorsements as of 2021, the highest globally (Government of Japan, 2021). Oshika et al. (2024) document substantial quality variation: large manufacturers provide detailed quantitative climate disclosures driven by EU Carbon Border Adjustment Mechanism exposure and sup-ply chain pressures, while financial and service firms rely on qualitative descriptions. Japanese corporations favor technological solution narratives – hydrogen energy, carbon capture – over explicit reduction commitments.

Social disclosure patterns reflect Japan's labor market challenges. Kato and Kodama (2018) examine how aging demographics (29.1% over age 65, globally highest) and workforce shortages shape ESG priorities. Although over 90% of Tokyo Stock Exchange (TSE) prime market firms set female manager targets, actual representation increased modestly from 3.7% (2015) to 6.2% (2020), substantially below the governmental 30% target for 2030. This 'commitment-action gap' motivates modality classification distinguishing stated plans from demonstrated outcomes.

Buchanan and Deakin (2024) demonstrate firms with higher outside director ratios and institutional ownership provide more detailed ESG disclosure aligning with performance. Japan's independent director system faces formalism criticism: many independents are retired executives or corporate affiliates, constraining governance oversight effectiveness.

Japanese ESG disclosure exhibits 'high quantity, moderate quality' characteristics: near-universal reporting among large corporations, yet variable depth and verifiability. Firms emphasize process inputs – establishing committees, conducting training – over measurable outcomes like emission reductions or audit results. This context renders action-level analysis particularly valuable for distinguishing substantive initiatives from symbolic compliance.

2.3. Knowledge graphs and LLM-driven text analysis

Systematic reviews map natural language processing contributions to sustainability research (Tashakori et al., 2025), documenting scholarly attention to ESG reporting, green innovation and circular economy themes. LLMs emerge as transformative tools for sustainability disclosure analysis (Mousavian Anaraki et al., 2025), enabling semantic extraction, automated classification, and greenwashing detection addressing traditional text mining limitations. These advances complement broader AI applications across business domains (Lakhal et al., 2025), demonstrating structured information extraction versatility.

Knowledge graphs structure information by organizing entities as nodes and relationships as edges within semantic networks (Huang et al., 2014; Loughran & McDonald, 2016). Unlike tabular relational databases, knowledge graphs enable multihop reasoning, path queries and community detection. Flexibility and interpretability constitute primary advantages: explicit relationship representation contrasts sharply with opaque 'black box' machine learning models.

Knowledge graph construction historically relied on manual curation (Freebase, DBpedia) and rule-based extraction utilizing regular expressions and dependency parsing. Supervised machine learning – Conditional Random Fields (CRF) and Support Vector Machines (SVM) for Named Entity Recognition and Relation Extraction – has been widely applied (Todd et al., 2024). Transformer-based models (BERT, RoBERTa) significantly improved NER and RE performance, yet require domain-specific fine-tuning using annotated datasets for optimal results.

LLMs transform knowledge extraction through in-context learning, instruction following, and semantic reasoning. Pan et al. (2024) identify three trends in LLM-driven knowledge graph development: zero-shot and few-shot extraction through prompt engineering achieves F1 scores of 0.85–0.92 without extensive labeled datasets; multi-hop reasoning and knowledge completion enables implicit relationship inference; automatic schema generation dynamically identifies entity and relationship types from text.

Kumar Gupta et al. (2024) demonstrate LLM-knowledge graph synergy for ESG analysis, extracting event triples from news supporting complex queries like 'which companies were fined for environmental violations'? Knowledge graph-enhanced systems improved accuracy by 23% over pure LLM approaches, validating structured knowledge value for ESG intelligence.

LLM deployment for ESG knowledge extraction faces three major challenges (Ji et al., 2023; Loughran & McDonald, 2016; Pan et al., 2024). Hallucination – generating plausible but non-existent information – poses significant risks when LLMs fabricate ESG actions or misattribute activities. Ji et al. (2023) identify 15–30% hallucination rates in domain-specific extraction without verification. Long document processing encounters computational constraints: corporate sustainability reports averaging 68 pages (range: 22–156) frequently exceed GPT- 4o's 128,000-token context window when including supplementary materials, risking critical disclosure omission through truncation. Entity standardization challenges arise from linguistic variation: LLMs may inconsistently extract 'employees', 'workforce', 'staff' and 'personnel' as distinct entities despite semantic equivalence, fragmenting network analysis.

The methodological framework systematically addresses these challenges. Hallucination mitigation requires mandatory evidence citation: each extracted action references specific source text passages with page numbers, enabling verification. Long document processing employs hierarchical chunking using 512-token segments with 50-token overlap, combined with retrieval-augmented generation (RAG) retrieving relevant chunks for disambiguation, preserving semantic continuity within context constraints. Entity standardization applies post-extraction clustering, leveraging semantic embeddings and human-validated canonical mappings consolidating synonymous terms while maintaining essential distinctions (e.g. 'board of directors' *versus* 'management' as distinct governance entities).

Network analysis methodologies – centrality, community detection, path analysis – constitute well-established social network research tools (Borgatti et al., 2009), yet remain underutilized for intra-organizational ESG action networks. Most ESG network studies examine inter-organizational networks (supply chains, interlocking directorates) rather than ESG topic interconnections within corporate disclosure narratives. This study addresses this gap by applying network analysis to topic co-occurrence and company similarity networks constructed from extracted knowledge graphs.

2.4. From aggregate to granular: positioning action-level extraction

This subsection positions action-level extraction methodology relative to existing ESG text mining approaches, clarifying incremental analytical granularity refinements and methodological shifts in answerable questions.

Recent work employs LLMs and RAG to extract structured ESG insights from sustainability reports, constructing knowledge graphs identifying disclosure similarities across regions and sectors (Bronzini et al., 2024). Such approaches demonstrate LLM feasibility for structured ESG extraction, yet most emphasize broad topical coverage over fine-grained action-level relationships required for stakeholder analysis and credibility assessment. Existing ESG text mining research varies in granularity, ranging from document-level topic modeling to sentence-level classification. Action-level extraction distinguishes itself from three dominant methods, each enabling different analytical capabilities.

2.4.1. Topic modeling (LDA/NMF): document-level thematic coverage

Traditional topic modeling treats documents as distributions over latent topics, identifying thematic prevalence without extracting structured relationships (Kiriu & Nozaki, 2020; Yoon et al., 2023). Latent Dirichlet Allocation (LDA) applied to Japanese ESG reports (Kiriu & Nozaki, 2020) reveals which sustainability themes companies emphasize but cannot answer relational queries: *Who reduced emissions? By how much? When? Through what specific actions?* This limitation restricts analysis to thematic coverage

assessment – measuring whether companies discuss diversity or climate – without capturing actionable intelligence required for stakeholder analysis or credibility evaluation.

2.4.2. Sentence-level classification (BERT-based): categorical labeling without relationships

Domain-specific language models classify sentences by ESG dimension or sentiment (Huang et al., 2023; Lee et al., 2025; Schimanski et al., 2024). Schimanski et al. (2024) develop NLP-based classifiers that quantify E, S, and G communication, while Huang et al. (2023) fine-tune FinBERT on 2000 manually annotated sentences to classify financial text into E/S/G/None categories. Despite greater granularity than document-level methods, these approaches lack *relational structure*. Sentences classified as 'positive environmental disclosure' reveal tone and topic but not actionable intelligence: whether actions constitute commitments or achievements, who performs them, or whom they benefit. Investor due diligence requires precisely these relational details – distinguishing 'we will reduce emissions' (future commitment) from 'employees reduced emissions 30% through solar panel installation' (completed action specifying actor, magnitude, mechanism).

2.4.3. Supervised named entity recognition and relation extraction (NER + RE): high accuracy, limited adaptability

Supervised deep learning models identify entities (companies, metrics, dates) and extract relationships (Lee & Kim, 2023; Zou et al., 2025), yet require *extensive domain-specific annotated training data* scarce for ESG disclosure. Heterogeneous reporting frameworks (GRI, SASB, TCFD) employ divergent terminologies and indicator sets; evolving taxonomies introduce emerging topics ('just transition', 'nature-related risks', 'AI ethics governance') lacking established annotation schemes; multilingual contexts present linguistic divergence as Japanese firms disclose in both Japanese and English with culture-specific framing. Fine-tuning BERT-based RE models for each framework-language-topic combination proves resource- prohibitive. Supervised models further struggle with *schema flexibility*: adding new entity types ('Scope 3 suppliers') or relationship types ('co-benefits to biodiversity') requires dataset re-annotation and model retraining, processes too slow for rapidly evolving ESG standards.

2.4.4. Action-level knowledge graphs: incremental granularity, methodological shift

This methodology represents *incremental innovation in granularity* and a *methodological shift in analytical paradigm*. Table 1 summarizes positioning relative to existing approaches.

2.4.4.1. Incremental innovation. Extraction decomposes each ESG disclosure into eight semantic elements (company, topic, actor, action verb, target, modality, evidence and confidence), yielding finer-grained structured records than sentence-level approaches. This granularity enables queries impossible with coarser methods: 'Which environmental actions explicitly benefit communities'? (requires actor–target-topic triangulation) or 'What proportion of diversity commitments transitioned to completed actions'? (requires modality-topic temporal analysis).

2.4.4.2. Methodological shift. Unlike supervised RE requiring domain-specific training data, LLM-based approaches achieve *zero-shot schema adaptation*. The 39-topic ESG taxonomy (Supplementary Appendix B) was defined without annotating training examples – GPT-4o's instruction-following capabilities enable extraction through taxonomy definitions and relationship schemas provided in natural language prompts. When new ESG issues emerge (biodiversity credits, AI ethics audits), taxonomy updates occur through prompt instruction modification rather than model retraining. This *schema flexibility* proves critical for ESG research, where disclosure standards and materiality definitions evolve faster than annotation pipelines adapt.

Table 1. Comparative positioning of ESG text mining approaches.

Method	Granularity	Relational structure	Schema adaptation	Training data
Topic modeling (LDA/NMF)	document	None (topic distributions)	fixed topics	unsupervised
Sentence classification (BERT)	Sentence	None(categorical labels)	Fixed categories	Supervised (labeled sentences)
Supervised NER + RE	Entity- relation pairs	Pairwise links	Requires re-training	Extensive annotations
This approach	Action-level (8 elements)	Multi-hop graph queries	Prompt- based (zero-shot)	Zero-shot

2.4.5. Three novel analytical capabilities

This extraction granularity and schema flexibility enable knowledge graph construction, which in turn unlocks three analytical capabilities unattainable with prior methods.

2.4.5.1. Stakeholder relationship mapping. Systematic actor–target pair extraction operationalizes stakeholder salience theory (Freeman, 1984; Mitchell et al., 1997) at scale, analyzing *who does what for whom*. Traditional methods cannot distinguish 'management discloses to investors' from 'employees implement initiatives for communities', yet these patterns reveal fundamentally different stakeholder prioritization. Analysis reveals 65.2% of disclosed actions lack specified actors (indicating organizational-level framing), while 8.2% explicitly attribute actions to employees (Section 4.1), enabling systematic examination of attribution patterns and stakeholder engagement rhetoric.

2.4.5.2. Credibility assessment via modality analysis. Distinguishing future commitments from ongoing actions from past achievements enables greenwashing detection unachievable through topic modeling or sentence classification. Commitment-heavy disclosure without supporting evidence of ongoing implementation or completed outcomes signals symbolic compliance rather than substantive action (Lyon et al., 2024). Modality distribution analysis (65.5% ongoing, 19.1% future commitments, 15.4% past achievements; Section 4.1) reveals disclosure rhetoric patterns: low future commitment ratios may reflect genuine operational focus or disclosure framework composition (sustainability reports emphasizing demonstrated actions *versus* integrated reports containing forward-looking projections).

2.4.5.3. Topic integration network analysis. Treating ESG topics as nodes and co-disclosure patterns as edges, knowledge graph structures enable network analysis revealing which topics firms conceptually bundle *versus* isolate. Emissions reduction frequently co-occurs with renewable energy and circular economy (forming environmental action clusters), while tax transparency rarely bridges to environmental topics (Section 4.2). This network perspective – enabled by structured relational extraction – exposes integration patterns invisible to bag-of-words topic models or sentence classifiers analyzing documents in isolation.

2.4.5.4. Graph-specific query capabilities. These capabilities uniquely depend on knowledge graph representation beyond LLM extraction alone. Two query types illustrate graph-specific value unachievable with tabular data:

Multi-hop reasoning traverses multiple relationship types answering complex questions. Query: 'Which companies disclose employee diversity training *and* those employees subsequently implement community engagement initiatives'? This requires traversing Company-DISCLOSES→Action(topic: diversity)-PERFORMED_BY→Actor(employees)-PERFORMS→Action(topic: community_engagement)-TARGETS→communities. Graph structure reveals 23 companies (16.3%) exhibiting this employee-mediated ESG pathway, operationalizing stakeholder theory's internal-external stakeholder linkage concept (Freeman, 1984). Tabular extraction cannot efficiently represent such multi-step relational patterns without extensive join operations and predetermined query templates.

Stakeholder-centric aggregation pivots analysis by actor or target rather than company. Query: 'What ESG actions do *suppliers* perform across all companies'? Graph traversal across Company-DISCLOSES→Action-PERFORMED_BY→Actor(suppliers) aggregates 1142 supplier-attributed actions (1.5% of total), revealing suppliers predominantly appear as *targets* of corporate actions (audits, guideline dissemination, 2.8%) rather than *actors* themselves. This asymmetry – visible through actor- target network analysis (Supplementary Appendix E) – exposes hierarchical supply chain governance contradicting partnership rhetoric. Such stakeholder-centric views require graph's flexible traversal capabilities; traditional document-company matrices fix analysis units at company level.

2.4.6. Accuracy-Scalability trade-offs

Supervised RE models fine-tuned on domain-specific corpora may achieve marginally higher precision ($F1 \approx 0.92–0.95$) than LLM approaches (validation accuracy 89.5%, Cohen's $\kappa = 0.95$) on narrowly defined entity types within specific reporting frameworks. LLM-based extraction scales across diverse frameworks (GRI-, SASB-, TCFD-aligned disclosures without separate models), multilingual contexts (Japanese and English reports processed uniformly), and evolving taxonomies (adding topics like nature-related financial risks

requires prompt updates, not dataset reannotation and retraining cycles spanning months). These scalability and flexibility advantages outweigh modest precision trade-offs for exploratory ESG research requiring comprehensive cross-framework coverage and rapid adaptation to emerging sustainability issues.

This positioning clarifies contributions extend beyond 'applying LLMs to ESG text'. The methodology establishes a fundamentally different analytical paradigm – shifting from thematic coverage assessment (what topics are discussed) to *traceable action intelligence and stakeholder relationship mapping* (who does what, for whom, when, with what evidence). This shift enables stakeholder theory operationalization, signaling credibility evaluation and network-based integration analysis at previously unattainable scales and granularity.

3. Data and methods

3.1. Analytical framework

This study employs a mixed-methods exploratory design combining AI-driven knowledge extraction with network analysis and theory-guided interpretation. The analytical framework comprises three integrated layers.

3.1.1. Data layer: from unstructured text to structured knowledge

Corporate sustainability narratives are transformed into machine-analyzable knowledge graphs through LLM-powered extraction. Each ESG action is represented as an eight-element record: (company, topic, actor, action, target, modality, evidence and confidence). This structured representation enables systematic pattern identification un-achievable through traditional qualitative document analysis while preserving action-level detail lost in aggregate ESG ratings.

3.1.2. Analysis layer: network and statistical methods

Two complementary networks are constructed from extracted actions. The *topic co-occurrence network* (nodes = ESG topics, edges = companies co-disclosing topic pairs) reveals integration patterns across ESG dimensions. The *company similarity network* (nodes = companies, edges weighted by cosine similarity of topic profiles) identifies strategic archetypes. Network analysis employs established metrics: density, clustering coefficient, modularity (community structure) and centrality (topic importance). Louvain clustering (Blondel et al., 2008) identifies company groupings, validated against industry classifications using Adjusted Rand Index (ARI). Theory layer: stakeholder, signaling and institutional perspectives. Empirical patterns are interpreted through three theoretical lenses (Connelly et al., 2011; DiMaggio & Powell, 1983; Freeman, 1984; Mitchell et al., 1997). Stakeholder theory guides actor–target relationship analysis, predicting prioritization patterns based on power, legitimacy and urgency attributes. Signaling theory informs modality distribution interpretation, distinguishing credible signals (ongoing actions with evidence) from cheap talk (future commitments without verification). Institutional theory explains archetype emergence and industry clustering as outcomes of coercive, mimetic and normative isomorphic pressures.

This framework positions the study as *knowledge graph-enabled exploratory research*, contributing methodological innovation (LLM-powered knowledge extraction and graph-based network analysis) and empirical insights (action-level disclosure patterns and stakeholder structures in Japanese ESG reporting). Knowledge graph approaches enable analyses impossible with traditional methods: systematic actor–target relationship mapping, ESG topic co-occurrence network construction and company archetype identification through graph clustering. The exploratory approach proves appropriate given limited prior research applying knowledge graphs to corporate ESG disclosure analysis, particularly in Japan's stakeholder capitalism context.

3.2. Data collection and setting

The sample comprises 141 Japanese corporations listed on the TSE Prime Market publishing standalone sustainability reports or integrated reports with ESG information for fiscal year 2024. Companies were

selected meeting four criteria: Prime Market listing, report availability in English or Japanese, report length exceeding 20 pages with significant ESG content, and document accessibility. The sample includes manufacturing (45%), financial services (18%), retail/consumer (15%), energy/utilities (12%) and other services (10%), representing approximately 62% of Prime Market capitalization.

Reports were collected from corporate investor relations or sustainability websites, averaging 68 pages (range: 22–156). Using PyMuPDF, text was extracted while filtering non-ESG content (covers, tables of contents, financial statements), removing headers/footers/page numbers, standardizing encoding to UTF-8, and validating extraction quality. Reports with >10% extraction errors underwent manual review and re-extraction.

3.3. ESG topic taxonomy development

The methodology relies on a 39-category ESG topic taxonomy (Supplementary Appendix B) structuring action classification and enabling systematic cross-company comparison. The taxonomy was developed using a hybrid deductive-inductive approach, integrating established frameworks with Japanese disclosure specificity.

3.3.1. Design rationale: balancing generalizability and specificity

This taxonomy balances generalizability (alignment with international standards) and specificity (Japanese disclosure patterns). While prior ESG disclosure studies predominantly relied on either universal frameworks like GRI (Marimon et al., 2012) or industry-specific standards like SASB (Khan et al., 2016), this hybrid approach addresses a methodological gap identified by recent research (Lin et al., 2024): the need for taxonomies capturing regional institutional variations while maintaining cross-national comparability. The 39-category structure enables both comparative analysis across markets and fine-grained examination of Japan-specific ESG patterns, particularly stakeholder-oriented disclosure themes underrepresented in Western materiality frameworks (Buchanan & Deakin, 2024; Mielcarz et al., 2023).

3.3.2. Framework integration

Three authoritative ESG disclosure standards were integrated: **Global Reporting Initiative (GRI) Standards** (Marimon et al., 2012) – the most widely adopted sustainability reporting framework defining material topics across ESG dimensions; **SASB Materiality Map** (Khan et al., 2016) – industry-specific disclosure topics emphasizing financially material ESG factors; **Task Force on Climate-related Financial Disclosures (TCFD) recommendations** (Bingler et al., 2022) – climate governance, strategy, risk management and metrics disclosure guidelines. GRI provides comprehensive stakeholder-oriented coverage, SASB ensures financial materiality relevance and TCFD addresses climate-specific detail – collectively covering corporate ESG disclosure breadth.

3.3.3. Japanese context adaptation

Topics prominent in Japanese sustainability reporting but underrepresented in Western standards were added to framework-derived categories. Prior research on Japanese corporate governance (Buchanan & Deakin, 2024; Kobayashi et al., 2024; Mielcarz et al., 2023) highlights recurring themes distinct from Western ESG priorities: *stakeholder dialogue* (nemawashi-oriented engagement), *social innovation* (Society 5.0 governmental initiatives) (Lechevalier, 2024) and *employee benefits* (lifetime employment tradition). These additions reflect Japan's stakeholder capitalism model (Miyajima & Saito, 2021) and align with national policy priorities.

3.3.4. Taxonomy structure

The final taxonomy contains 39 topics organized into three pillars: Environmental (14 topics), social (15 topics) and governance (10 topics). Environmental topics span climate (emissions reduction, renewable energy, carbon pricing), resource management (water, waste, circular economy) and ecosystem protection (biodiversity, pollution control). Social topics address human capital (diversity, talent development, occupational safety), stakeholder relations (community engagement, supply chain labor, indigenous

rights) and consumer protection (product safety, customer privacy). Governance topics cover board structure, executive compensation, risk management, compliance, anti-corruption and transparency mechanisms (tax, data security). Each topic is defined with inclusion/exclusion criteria and example disclosures to ensure consistent LLM classification. Supplementary Appendix B provides complete topic definitions and ESG category assignments.

3.3.5. Taxonomy construction methodology

Developing a unified taxonomy from three established frameworks (GRI, SASB, TCFD) while preserving Japanese disclosure specificity presented three methodological challenges: resolving framework overlaps, classifying boundary cases and determining optimal granularity. Each challenge is addressed systematically below.

Resolving Framework Overlaps. The methodology applied three resolution principles: (1) Temporal scope differentiation *via* modality – when frameworks address the same ESG dimension but differ in temporal orientation(e.g. GRI 305 current emissions *versus* TCFD forward-looking climate plans (Bingler et al., 2022)), they were unified under single topics(e.g. emissions reduction) and distinguished temporal framing through modality fields ('achieved 20% reduction' = past_achievement; 'committed to net-zero by 2050' = fu- ture_commitment). This preserves coherence while enabling credibility assessment (Section 4.3). (2) Granularity consolidation with action-level specificity – when overlaps reflected different granularity(e.g. GRI 404 'Training' *versus* SASB 'Employee Development'), the approach adopted broader categories (talent development and training) and captured distinctions through action fields ('provided 40 hours annual training' *versus* 'implemented mentorship program'), avoiding taxonomy fragmentation. (3) Materiality-based filtering – retaining only ESG-material topics based on stakeholder salience (Freeman, 1984) and Japanese disclosure prevalence (e.g. tax transparency, economic value distribution), excluding general financial performance).

3.3.5.1. Classifying ambiguous boundary cases. Multi-topic actions required primary-topic classification rules with network-based co-occurrence capture. Primary topic assignment follows the action's direct object: 'Installing solar panels to reduce Scope 2 emissions by 15%' is classified under renewable energy adoption(primary action), not emissions reduction. Co-occurrence networks (Section 4.4) subsequently capture the renewable energy–emissions reduction linkage through co-disclosure analysis (73% of firms; Table 2). This separates *classification* (unique assignment) from *integration analysis* (cross-topic relationships), preserving multi-topic connections without artificial categorical boundaries. Similarly, 'Diversity training for supply chain partners' is assigned to diversity and inclusion, while Actor/Target fields capture stakeholder relationships.

Rationale for 39 Topics: Balancing Framework Coverage and Classification Consistency. The final count emerged from framework integration logic balanced against granularity constraints: Theoretical lower bound (~25–30 topics) – excessive consolidation risks losing policy-relevant distinctions (merging *water stewardship* and *waste management* conflates water scarcity *versus* circular economy, governed by distinct Japanese regulations (Kobayashi et al., 2024). Theoretical upper bound(~50+ topics) – excessive fragmentation introduces classification inconsistency (splitting *emissions reduction* into Scope 1/2/3 creates high semantic overlap and misclassification risk, especially when Japanese reports discuss emissions holistically; Hogan et al., 2022). Framework-driven convergence to 39 – our taxonomy contains 14 Environmental topics (TCFD pillars + GRI biodiversity/pollution), 15 Social topics (Japanese

Table 2. Standardized ESG action disclosure template.

Topic	Actor	Action description	Modality	Evidence/metric
Emissions reduction	Company	Installed solar panels at 15 facilities	Past (2023)	15% Scope 2 reduction
Diversity and inclusion	Board of directors	Committed to 30% female director quota	Future (2030 target)	Board resolution May 2024
Supplychain governance	Management	Conducting annual supplier ESG audits	Ongoing	120 suppliers audited in 2023

Example format for structured ESG reporting based on the 8-element extraction schema, enabling cross-firm comparability and automated compliance monitoring.

stakeholder capitalism emphasis; Miyajima & Saito, 2021; Buchanan & Deakin, 2024) and 10 Governance topics (GRI 2–9 to 2–27; Marimon et al., 2012), balancing international framework comprehensiveness with Japanese institutional specificity (Mielcarz et al., 2023). Each topic has defined inclusion/exclusion criteria (Supplementary Appendix B). Empirical analysis of co-disclosure patterns reveals high cross-dimensional integration: topics from different ESG dimensions co-occur 1.18× more frequently than topics within the same dimension (between-dimension mean = 35.3, within-dimension mean = 29.8 companies per topic pair). This pattern reflects Japanese firms' integrated ESG strategies that deliberately link ESG initiatives (Buchanan & Deakin, 2024; Miyajima & Saito, 2021), consistent with stakeholder capitalism models. While the taxonomy maintains normative alignment with international frameworks (GRI, SASB, TCFD), the high integration validates our network-based analytical approach that captures cross-dimensional relationships rather than imposing artificial categorical boundaries. A methodological caveat warrants acknowledgement: because both the 39-topic taxonomy and the sampled firms' disclosure practices are grounded in the same underlying frameworks (GRI, SASB, TCFD), the observed convergence between taxonomy structure and co-disclosure patterns is partially expected rather than fully independent evidence of construct validity. In other words, firms reporting under GRI or TCFD guidance are predisposed to disclose topics already captured by a taxonomy built from those same standards, which may inflate apparent coherence. External validation against independently derived classification schemes – such as inductive content-analysis taxonomies developed without reference to GRI, SASB or TCFD – would provide a more rigorous test of the taxonomy's empirical grounding and is recommended for future work.

3.4. LLM-driven ESG action extraction

This study developed a three-stage LLM-driven extraction pipeline to convert unstructured sustainability narratives into structured action records. Each record contains eight elements: (company, topic, actor, action, target, modality, evidence and confidence). The framework draws on theory, where Actor/Target extraction is based on stakeholder salience theory (Freeman, 1984; Mitchell et al., 1997), and Modality extraction is based on the credibility dimension of signaling theory (Connelly et al., 2011). The detailed Topic classification enables network analysis of ESG integration patterns. Figure 1 provides an overview of the complete pipeline from data collection to network construction.

3.4.1. Stage 1: Weak extraction (broad coverage)

This study used GPT-4o as the primary extraction model (Pan et al., 2024), segmenting reports into 500-word chunks with 50-word overlap. For each chunk, the LLM extracted all ESG-related actions into eight-element records: (1) assignment to one of 39 ESG topics (Supplementary Appendix B), (2) actor categorization (company, employees, board, suppliers, government, NGO, other stakeholders), (3) action verbs from a controlled vocabulary, (4) target beneficiaries, (5) modality classification (past_action, ongoing, future_commitment), (6) verbatim evidence text and (7–8) confidence scores (0–1). The prompt design (Supplementary Appendix A) prioritized recall over precision, generating approximately 92,000 raw records including duplicates and low-confidence extractions, ensuring comprehensive coverage of disclosed actions.

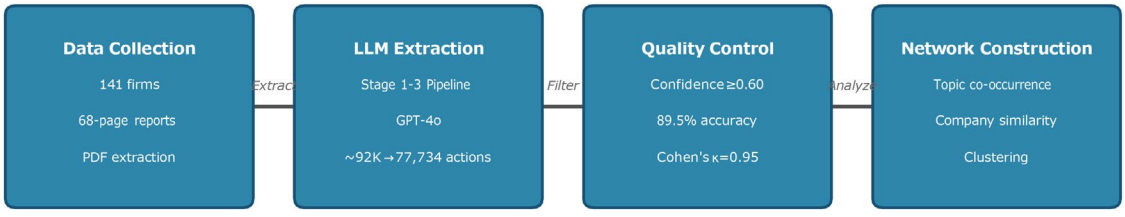


Figure 1. Four-stage ESG knowledge graph construction pipeline.

3.4.2. Confidence score calculation

To ensure extraction quality, we implemented a confidence assessment combining four dimensions informed by RAG system evaluation (Saad-Falcon et al., 2024; Salemi & Zamani, 2024): (1) *Evidence consistency* (weight: 0.4) – verifying extracted evidence exists verbatim in source documents through exact string matching; (2) *Topic alignment*(weight: 0.2) – measuring semantic similarity between extracted topics and our 39-category taxonomy using sentence embeddings (cosine threshold ≥0.85); (3) *Action specificity* (weight: 0.2) – assessing whether action verbs are concrete (e.g. 'installed', 'reduced') *versus* abstract (e.g. 'addressed', 'considered'); (4) *Retrieval quality* (weight: 0.2) – evaluating RAG context relevance for disambiguation cases. The weighted combination (detailed formula in Supplementary Appendix D.2) yields final confidence scores, with records below 0.60 filtered out. This threshold balances precision and recall, resulting in 77,734 records with 97.7% exceeding 0.90 confidence, indicating high extraction reliability.

3.4.3. Stage 2: Strong verification (quality refinement)

Raw records underwent three verification sub- processes:(1) *Deduplication* – computing semantic similarity between action descriptions using sentence embeddings (all-MiniLM-L6-v2), clustering records with cosine similarity >0.85, and retaining highest- confidence instances per cluster(11.3% reduction);(2) *Confidence filtering* – excluding records below 0.60 threshold (3.8% reduction), validated through manual evaluation confirming monotonic accuracy increases with confidence (77.5% for 0.60–0.69, 89.5% for 0.80–0.89, 97.7% for 0.90–1.0; Supplementary Appendix C); (3) *Entity standardization* – hierarchical clustering of actor/target labels to consolidate equivalent terms('workers', 'staff', 'employees' → 'employees') and prevent network fragmentation.

3.4.4. Stage 3: RAG-enhanced contextual validation

For ambiguous records(confidence 0.60–0.74 or pronoun-referenced entities), we employed a three-layer RAG architecture (Supplementary Appendix D.1): (1) *Local context retrieval* – proximate text segments (200–300 words) preceding/succeeding evidence spans to capture immediate narrative context; (2) *Document-level retrieval* – semantic search across full reports using dense passage retrieval (top-3 similar passages via cosine similarity) to resolve cross-references; (3) *Graph-level retrieval* – querying partially constructed knowledge graphs for previously extracted actions from the same company to disambiguate entity references. Combined context was provided to GPT-4o for re-extraction, increasing accuracy of ambiguous records from 82.1% to 91.7%. Unresolved records (1.2%) were excluded.

The final dataset comprised 77,734 high-confidence records (confidence ≥ 0.60), with 97.7% exceeding 0.90 confidence and mean confidence 0.994.

3.4.5. Independent validation with human coders

To assess extraction reliability beyond model confidence scores, we conducted independent human validation on 100 stratified random samples reflecting the actual confidence distribution. Two independent validators, both holding PhDs with ESG research expertise, assessed whether all eight record elements (company, topic, actor, action, target, modality, evidence and confidence) accurately reflected the source text. Inter-rater reliability reached Cohen's $\kappa = 0.95$ with 99% consensus rate. The validation yielded 89.5% average accuracy, confirming that high-confidence extractions (0.80+) reliably capture disclosed actions. Detailed validation procedures and error analysis are provided in Supplementary Appendix C.

3.5. Knowledge graph construction

The framework transforms extracted action records into a property graph stored in Neo4j (Angles & Gutierrez, 2008), a native graph database optimized for relationship-intensive queries. This schema design balances expressiveness (capturing ESG action semantics) with query efficiency(enabling network analysis).

3.5.1. Graph schema design

The knowledge graph implements a multi-relational schema with five node types and four edge types. *Node types*: (1) Company (n = 141) – properties: name, industry_sector, market_cap_tier; (2) Topic (n = 39) – properties: name, ESG_pillar (E/S/G), definition; (3) Action (n = 77,734) – properties: action_verb, modality (past/ongoing/future), evidence_text, evidence_page, confidence_score, extraction_date; (4) Actor (n = 147 unique) – properties: actor_type (company/employee/board/supplier/NGO/government standardized_name; (5) Target (n = 198 unique) – properties: target_type (employee/community/environment/product/gov standardized_name). The 147 unique actors were determined through hierarchical clustering of raw actor labels using semantic embeddings (all-MiniLM-L6-v2), consolidating synonymous terms (e.g. 'workers', 'staff', 'employees' merged into canonical 'employees') while preserving meaningful distinctions (e.g. 'board of directors' *versus* 'management' as separate governance entities). This standardization process balanced network analysis requirements (avoiding excessive fragmentation) with organizational reality (maintaining substantive actor-type differentiation).

Edge types encode semantic relationships:

- Company-DISCLOSES→Action – Represents disclosure authorship. Edge properties: report_year, report_type(standalone/integrated).
- Action-BELONGS_TO→Topic – Classifies actions into ESG categories. Allows multiple topics per action for integrated initiatives.
- Action-PERFORMED_BY→Actor – Identifies action initiator. Captures governance structure(who acts).
- Action-TARGETS→Target – Identifies action beneficiary/object. Captures stakeholder focus (who benefits).

3.5.2. Graph database implementation

The extracted action records were instantiated as a property graph in Neo4j (Angles & Gutierrez, 2008), yielding 92,000 nodes and 233,000 edges from 77,734 action records. To ensure efficient traversal for network analysis, we implemented three indexing strategies:(1) B-tree indexes on Company.name and Topic. name for O(log n) entity lookup;(2) full-text indexes on Action.evidence_text enabling semantic search over source citations;(3) composite indexes on (Action.confidence_score, Action.modality) accelerating filtered queries by quality and temporal frame. The resulting graph structure exhibits moderate connectivity (average node degree = 5.1, diameter = 6), suitable for community detection and path-based analyses.

3.5.3. Representational affordances of graph structures

Relative to relational schemas, the graph model provides three analytical advantages for ESG action data. First, multi-topic actions (e.g. 'sup- plier diversity training' addressing both diversity and supply chain topics) are represented as multiple BELONGS_TO edges rather than requiring junction tables and join operations, preserving semantic clarity. Second, stakeholder pathways (e.g. tracing actions from board decisions to employee outcomes) are expressed as graph traversals rather than recursive common table expressions, improving query expressiveness. Third, provenance metadata (evidence citations, confidence scores) are preserved as edge properties, enabling filtered analysis without schema denormalization.

This graph structure enables network analyses: topic co-occurrence networks derive from shared Company-DISCLOSES→Action-BELONGS_TO→Topic paths; company similarity networks compute from Topic distribution vectors aggregated across actions. Neo4j's Cypher query language facilitates these complex pattern-matching operations (Francis et al., 2018).

3.6. Analytical methods

This study conducted three analytical workflows to analyze the knowledge graph: (1) descriptive statistics characterizing disclosure distributions, (2) network analysis revealing topic co-occurrence and company similarity patterns and (3) validation procedures assessing extraction reliability.

3.6.1. Descriptive analysis

Frequency distributions were calculated across five dimensions to identify disclosure trends: (1) *Topic distribution* – quantifying actions per ESG topic to identify disclosure priorities and the balance among dimensions (E-S-G); (2) *Action distribution* – aggregating types of action verbs to characterize the rhetoric of disclosure, where a high frequency of 'disclose' may imply an emphasis on transparency, whereas a high frequency of 'reduce' suggests a focus on operational change; (3) *Actor distribution* – identifying entities responsible for disclosed actions to map structures of organizational responsibility; (4) *Target distribution* – identifying beneficiaries of actions to uncover patterns of stakeholder salience (Freeman, 1984; Mitchell et al., 1997); (5) *Modality distribution* – comparing frequencies of past actions, ongoing activities, and future commitments to evaluate the credibility signaling of disclosures (Connelly et al., 2011). These distributions were examined both overall and disaggregated by ESG dimension (E *versus* S *versus* G) and industry sector to detect systematic variations.

3.6.2. Network construction and analysis

This analysis constructed two projection networks from the knowledge graph:

1. *Topic Co-occurrence Network*: An undirected weighted network where nodes represent ESG topics (n = 39) and edges connect topics co-disclosed by the same company. Edge weight equals the number of companies co-disclosing both topics. This network reveals ESG integration patterns – tightly connected topics indicate bundled disclosure strategies; isolated topics suggest siloed approaches. The methodology applied the Louvain community detection algorithm (Blondel et al., 2008) to identify topic clusters. Louvain iteratively maximizes modularity (Q), measuring intra-cluster edge density relative to random expectation. Modularity ranges from −0.5 to +1.0; $Q > 0.3$ indicates meaningful community structure. This analysis computed network-level metrics:(a) *density* (proportion of observed edges to possible edges), (b) *average clustering coefficient* (extent to which topics' neighbors are also connected, indicating triadic closure),(c) *modularity* (strength of community boundaries).
2. *Company Similarity Network*: Each company was represented as each company as a 39-dimensional vector where element i equals the count of actions disclosed for topic *i*. This analysis computed cosine similarity between all company pairs and constructed a network where nodes represent companies (n=141) and edges connect companies with similarity > 0.70 (empirically chosen threshold yielding interpretable clusters). The methodology applied Louvain clustering to identify company archetypes – groups sharing similar ESG disclosure profiles.

To validate archetype stability and assess whether ESG disclosure patterns align with industry structure, we computed the ARI comparing Louvain-derived clusters to industry sector classifications. Industry classifications were obtained from two sources: (1) TSE 33-sector classification system (Campbell et al., 2014) for companies listed on Japanese exchanges (n = 128); (2) Global Industry Classification Standard (GICS) (MSCI Inc. & S&P Dow Jones Indices, 2024) 11-sector taxonomy for companies with international listings or reporting (n = 13). We manually mapped TSE sectors to GICS sectors to create a unified 11-category classification (Supplementary Appendix G provides detailed mapping table). The mapping process consolidated fine-grained TSE categories (e.g. 'Electric Power' and 'Gas') into broader GICS equivalents ('Utilities'), and aligned Japan-specific sectors (e.g. 'Marine Transportation', 'Land Transportation', 'Air Transportation') to 'Industrials'. The 11 sectors used for ARI calculation are: Energy, Materials, Industrials, Consumer Discretionary, Consumer Staples, Health Care, Financials, Information Technology, Communication Services, Utilities and Real Estate.

ARI measures agreement between two clustering solutions, adjusting for chance. It ranges from −0.5 (worse than random) to +1.0 (perfect agreement), with 0 indicating random-level agreement. ARI is computed as:

$$ARI = \frac{\sum_{ij} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] / \binom{n}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] / \binom{n}{2}} \quad (1)$$

where n_{ij} is the number of companies assigned to both Louvain cluster i and industry sector j, a_i is the number of companies in Louvain cluster i, and b_j is the number in industry sector j. ARI = 1 indicates perfect agreement (Louvain clusters match industry sectors exactly), ARI = 0 indicates random assignment, and negative values indicate systematic disagreement. This analysis computed ARI using the scikit-learn implementation (Pedregosa et al., 2011).

4. Results

4.1. Descriptive patterns

Table 3 summarizes knowledge graph statistics. The 141 companies disclosed an average of 551.3 ESG actions (SD = 318.7, range: 112–1,765). Action distribution is right-skewed: the top 10% of companies (n = 14) account for 28.6% of all actions, while the bottom 25% (n = 35) contribute only 7.1%.

Contrary to assumptions that environmental issues dominate ESG disclosure, dimensional analysis reveals a social emphasis: Social(S) topics comprise 55.4% of actions, environmental (E) 26.9% and governance (G) 17.7% (Figure 2). This social dominance reflects Japan's labor market challenges and stakeholder capitalism orientation (Buchanan & Deakin, 2024; Kato & Kodama, 2018). Within environment, climate-related topics (emissions, renewable energy, climate adaptation) constitute 21.5% of all actions, demonstrating TCFD influence (Government of Japan, 2021). At the topic level, disclosure is highly concentrated (Figure 4; Table 6 in Supplementary Appendix E). Emissions reduction is the single most disclosed topic (15.2%), followed by diversity and inclusion (12.6%). Together, the top two topics account

Table 3. Knowledge graph statistics (n = 141 companies).

Metric	Value
Total ESG action records	77,734
Average actions per company	551.3 (SD = 318.7)
Unique ESG topics	39
Unique standardized actions	3,256
Unique actor types	147
Unique target types	198
Average extraction confidence	0.994
Proportion high confidence(≥ 0.9)	97.7%

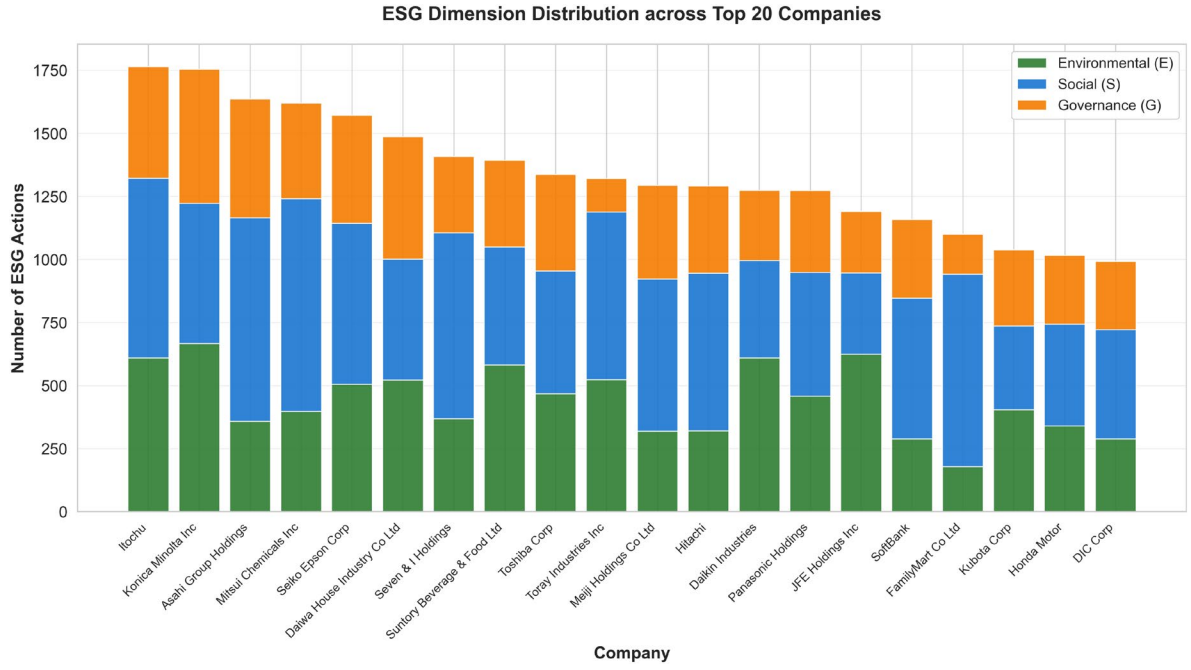


Figure 2. ESG dimensional distribution. Social topics dominate (55.4%), followed by environmental (26.9%) and governance (17.7%), challenging assumptions of environmental primacy in Japanese ESG disclosure.

for over one-quarter (27.8%) of all actions, and the top 15 topics represent 73% of disclosure. This concentration suggests Japanese firms focus ESG efforts on a relatively narrow set of high-salience issues rather than pursuing comprehensive coverage across all 39 topics in the taxonomy (Supplementary Appendix B).

4.1.1. Action type distribution

Analyzing action verbs reveals disclosure rhetoric patterns that reflect the distinction between symbolic and substantive ESG engagement (Lyon & Maxwell, 2011) (Table 7 in Supplementary Appendix E). The two most frequent actions – 'establish'(13.6%) and 'disclose' (13.5%) – dominate, together accounting for 27.1% of all disclosed actions. 'Establish' signals structural changes (establishing committees, policies, systems), representing what institutional theory terms *structural isomorphism* – adopting visible governance forms that signal legitimacy (DiMaggio & Powell, 1983). 'Disclose' emphasizes transparency and information provision, aligning with accountability-oriented disclosure regimes (Krueger & Sautner, 2024). The third and fourth most common actions – 'implement' (9.8%) and 'install' (9.0%) – represent operational execution, suggesting movement beyond purely symbolic commitments toward enacted practices.

This distribution suggests Japanese firms balance institutional signaling (establishing governance structures) with operational implementation, though the high 'disclose' frequency may reflect compliance- driven transparency rather than voluntary substantive action. To interpret these patterns, this analysis applies the conceptual framework from greenwashing literature (Lyon et al., 2024), which distinguishes three levels of corporate ESG engagement: *talk* (commitments, policies and disclosure statements), *decisions* (resource allocation and structural changes) and *action* (implemented practices with measurable outcomes).

Applying this framework to the action verb distribution reveals a hierarchical pattern. 'Disclose' (13.5%) represents pure *talk* – communicating ESG commitments without necessarily implementing them. 'Establish' (13.6%) straddles *talk* and *decisions* – creating governance structures (e.g. sustainability committees, ethical codes) signals commitment but does not guarantee operational change. 'Implement'(9.8%) and 'install'(9.0%) represent *decisions* transitioning to *action* – allocating resources to execute ESG programs. However, outcome-oriented verbs like 'reduce' (5.6%) and 'achieve'(3.2%) – representing completed *actions* with verified results – rank significantly lower.

The dominance of talk-oriented actions ('disclose' + 'establish' = 27.1%) over outcome-oriented actions ('reduce' + 'achieve' = 8.8%) may indicate decoupling between stated commitments and demonstrated results. This pattern aligns with institutional theory's observation that organizations adopt symbolic structures to gain legitimacy while potentially decoupling formal policies from actual operations (DiMaggio & Powell, 1983). However, the substantial presence of implementation verbs ('implement', 'install', 'conduct') suggests Japanese firms are not purely engaged in symbolic compliance but are taking intermediate steps toward operationalizing ESG commitments.

Notably, outcome-oriented actions like 'reduce'(5.6%) and 'achieve'(3.2%) rank lower, indicating that firms more frequently report process inputs (training, auditing and monitoring) than measurable outputs (actual reductions, certifications attained). This input–output imbalance aligns with two explanations grounded in institutional context and strategic disclosure theory.

First, Japan's corporate disclosure culture has historically emphasized process quality and continuous improvement (*kaizen*) over quantified performance outcomes (Kato & Kodama, 2018). Unlike Anglo-American disclosure regimes that prioritize measurable KPIs and outcome metrics (Krueger & Sautner, 2024), Japanese firms tend to report systematic processes, governance structures and procedural rigor as signals of organizational capability. This process orientation may lead firms to disclose controllable activities (conducting training, establishing monitoring systems) whose implementation can be verified internally, rather than uncertain outcomes (emission reductions, diversity targets) that depend on external factors and long-term horizons. Second, strategic ambiguity theory suggests that firms facing institutional pressures but uncertain about ESG effectiveness may adopt deliberate vagueness in outcome disclosure (Lyon et al., 2024). Disclosing inputs allows firms to demonstrate ESG engagement without committing to specific performance benchmarks that could invite scrutiny if unmet. Recent evidence on mandatory ESG disclosure regimes (Krueger & Sautner, 2024) shows that firms respond to compliance pressures by expanding process-oriented disclosure while maintaining strategic flexibility on outcome commitments – a pattern consistent with these findings.

The prevalence of 'train' actions (7.8% for general training; 2.7% specifically for safety training) underscores human capital development as a core ESG strategy, consistent with Japan's lifelong employment traditions (Kato & Kodama, 2018; Kobayashi et al., 2024) and knowledge-intensive economy. Training represents a controllable input with immediate documentation (number of sessions, participants) but deferred outcomes(improved employee wellbeing, productivity gains), fitting the input-disclosure preference pattern.

4.1.2. Actor distribution

Most disclosed actions (approximately 83%) do not specify granular actors, instead attributing actions generically to 'the company' or organizational units. Among the 17% specifying actors, the distribution reveals organizational ESG responsibility structures. 'Employees' appear as actors in 8.2% of cases, 'board of directors' in 4.3%, 'suppliers' in 2.1% and 'management' in 1.7%. This pattern suggests ESG actions are framed as collective organizational initiatives rather than attributable to specific internal or external stakeholders. The relatively low supplier actor frequency (2.1%) indicates supply chain ESG remains underdeveloped – firms disclose their own actions toward suppliers more than suppliers' ESG actions, pointing to hierarchical rather than collaborative supply chain governance.

The prominence of employee actors (8.2%) in a context where most actions lack specified actors is significant. When firms do specify actors, they emphasize employee-driven initiatives (grassroots engagement, bottom-up innovation), which may suggest inclusive ESG culture. However, this pattern may also reflect attribution bias – firms highlight employee contributions to burnish employer brand reputation while keeping negative outcomes organizationally diffuse.

4.1.3. Target distribution

Stakeholder targeting patterns reveal prioritization hierarchies with stark con- centration (Table 8 in Supplementary Appendix E; Figure 2). Employees are by far the dominant beneficiaries, appearing as targets in 10.2% of actions – 13 times more frequent than the next category (suppliers at 1.5%). This extreme employee-centricity reflects Japan's stakeholder capitalism tradition prioritizing employment security and worker welfare (Buchanan & Deakin, 2024; Freeman, 1984). Actions targeting employees span diversity training, wellbeing programs, safety improvements, and skill development, operationalizing stakeholder salience theory's prediction that powerful, legitimate, and proximate stakeholders(employees) receive disproportionate attention (Mitchell et al., 1997).

The stakeholder target heatmap (Figure 3) visualizes the concentration of actor–target relationships across the top 20 companies, revealing systematic employee-centric patterns. The intensity of employee-focused actions is evident across all major firms, with minimal engagement directed toward external stakeholders such as communities, customers or NGOs.

External stakeholder targets receive far less emphasis: local communities (0.5%), customers (0.3%) and society broadly (0.2%) together account for only 1% of targeted actions. This pattern may challenge narratives of broad stakeholder capitalism – in practice, Japanese ESG disclosure exhibits insider (employee) orientation over outsider (community, NGO, customer) engagement. Suppliers and supply chains collectively appear as targets in 2.8% of actions, primarily in contexts of auditing, training and guideline dissemination, suggesting supply chain ESG may balance compliance-driven and partnership-based approaches (Nguyen & Zuidwijk, 2025).

Interestingly, environmental targets like 'CO_2 emissions' (0.8%) and 'GHG emissions' (0.6%) are explicitly named, reflecting quantification and measurement norms in environmental disclosure. In contrast, social targets are more diffuse ('employees', 'workplace environment'), and governance targets often abstract ('stakeholders', 'entire company'), suggesting differential precision standards across ESG dimensions.

4.1.4. Modality distribution

The temporal framing of disclosed actions reveals credibility signals (Figure 4). Ongoing actions dominate at 65.5%, vastly exceeding future commitments (19.1%) and past actions (15.4%). This pattern appears inconsistent with greenwashing critiques emphasizing empty pledges – Japanese firms predominantly disclose continuous, operational ESG activities rather than aspirational future goals.

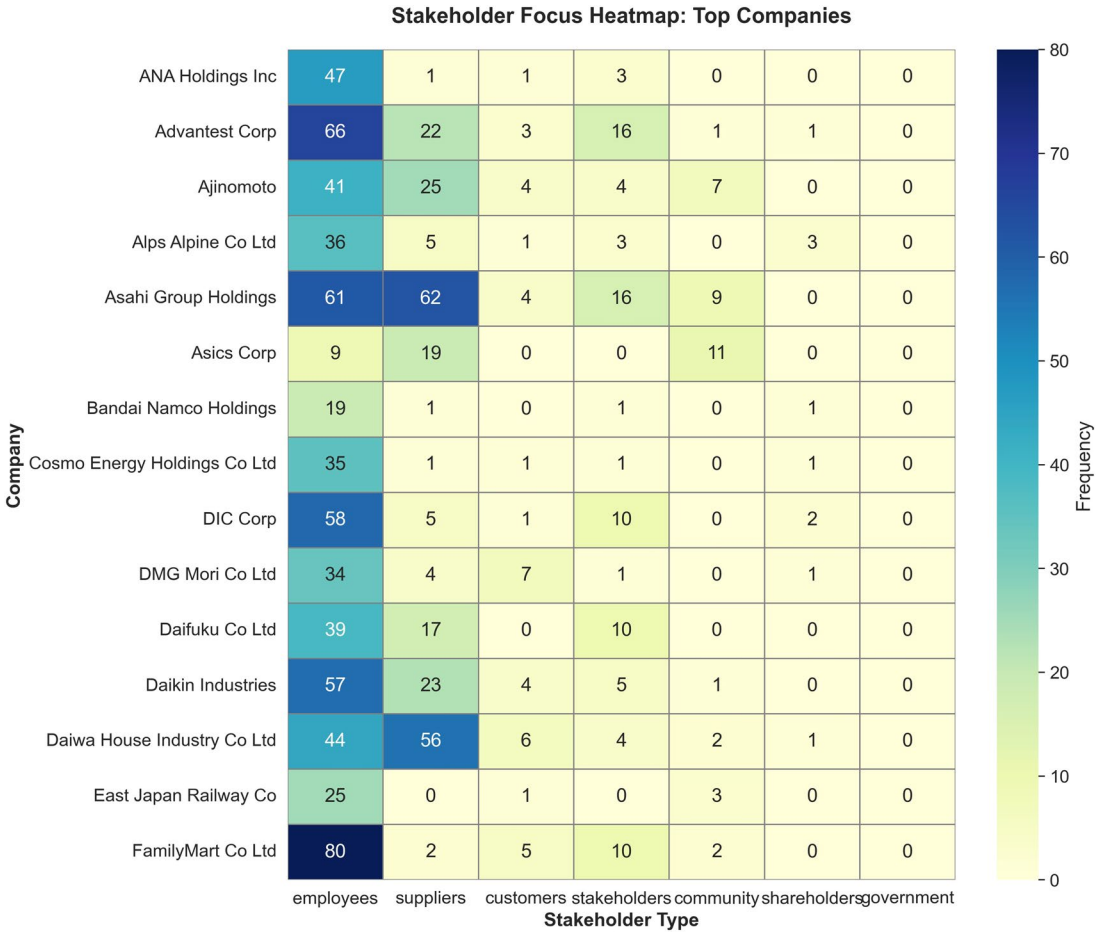


Figure 3. Stakeholder target heatmap. Matrix showing frequency of actor–target relationships across top 20 companies and stakeholder types, revealing employee-centric disclosure patterns.

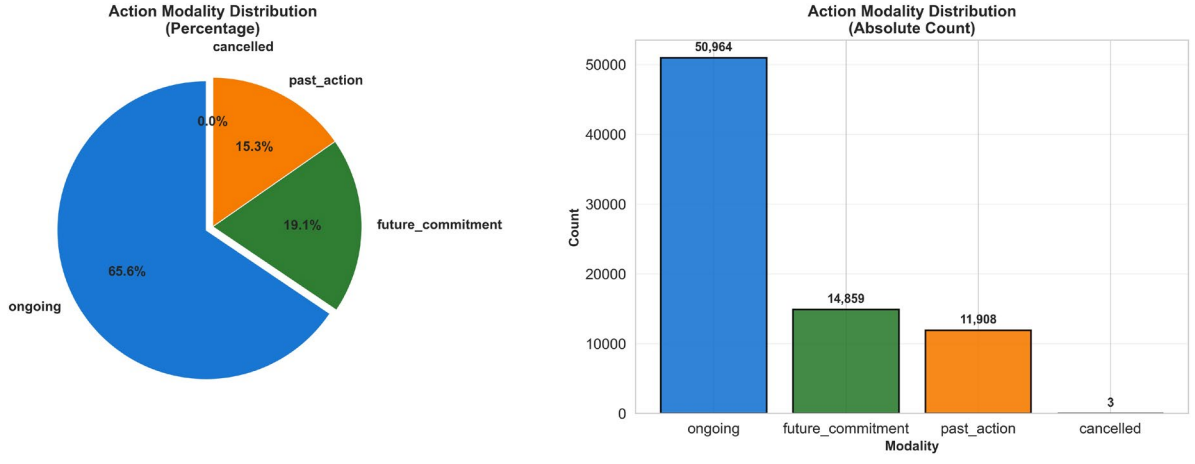


Figure 4. Action modality distribution across ESG dimensions. The prevalence of ongoing actions (65.5%) over future commitments (19.1%) suggests action-oriented disclosure. Environmental actions show higher future commitment ratios, indicating transition planning.

From a signaling theory perspective (Connelly et al., 2011), ongoing actions are moderately credible signals: harder to fabricate than mere commitments, but not independently verifiable like completed past actions with measurable outcomes.

The low proportion of past completed actions(15.4%) may reflect two dynamics:(1) Recency bias – firms emphasize current-year activities over historical achievements; (2) Outcome uncertainty – firms may hesitate to claim 'completed' status for complex ESG transitions (e.g. diversity transformation, emission reductions) requiring sustained multi-year efforts. Future commitments, while lower than ongoing actions, still comprise nearly one-fifth of disclosure, concentrated in environmental topics (especially carbon neutrality targets for 2030/2050). This environmental commitment intensity likely reflects regulatory pressures (TCFD, carbon pricing anticipation) and industry decarbonization roadmaps requiring long-term investment horizons.

Cross-dimensional comparison reveals environmental actions exhibit the highest future commitment ratio (28.3%) *versus* social (16.1%) and governance (12.8%), confirming that climate transition is framed as forward-looking strategic planning, whereas social and governance ESG are presented as embedded operational practices. This temporal dimension differentiation has implications for investor assessment – environmental ESG may require patience and commitment tracking, while social/governance ESG can be evaluated through current operational evidence.

4.2. Network structure

4.2.1. Topic co-occurrence network: high integration and triadic structure

The topic co-occurrence network exhibits high density (0.68), indicating most ESG topics are co-disclosed rather than isolated (Figure 6; Table 4). Of the 39 = 741 possible topic pairs, 504 (68%) appear together in at least one company's disclosure, suggesting firms adopt integrated ESG strategies spanning multiple topics rather than narrowly focused single-topic reporting.

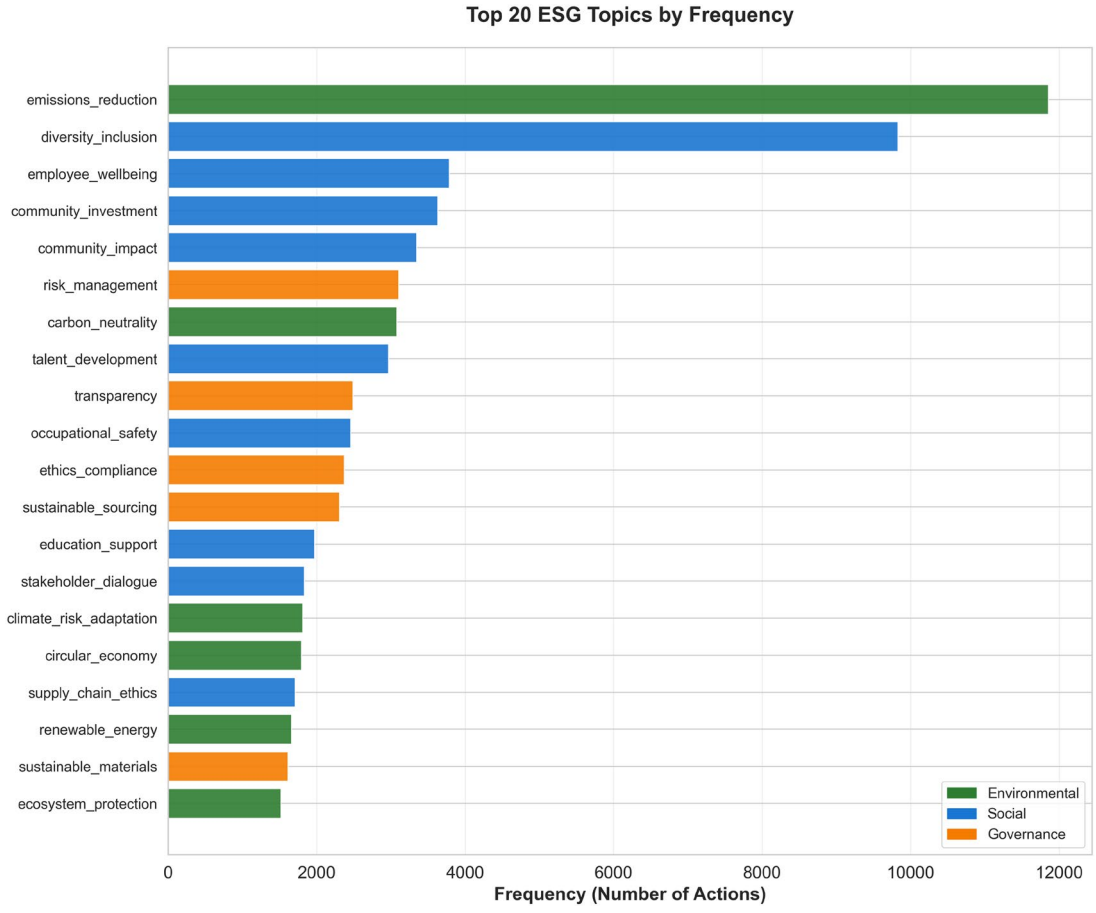


Figure 5. Top 20 ESG topics by action frequency. The distribution shows strong concentration in emissions reduction (15.2%) and diversity and inclusion(12.6%), with the top 15 topics accounting for 73% of all disclosed actions.

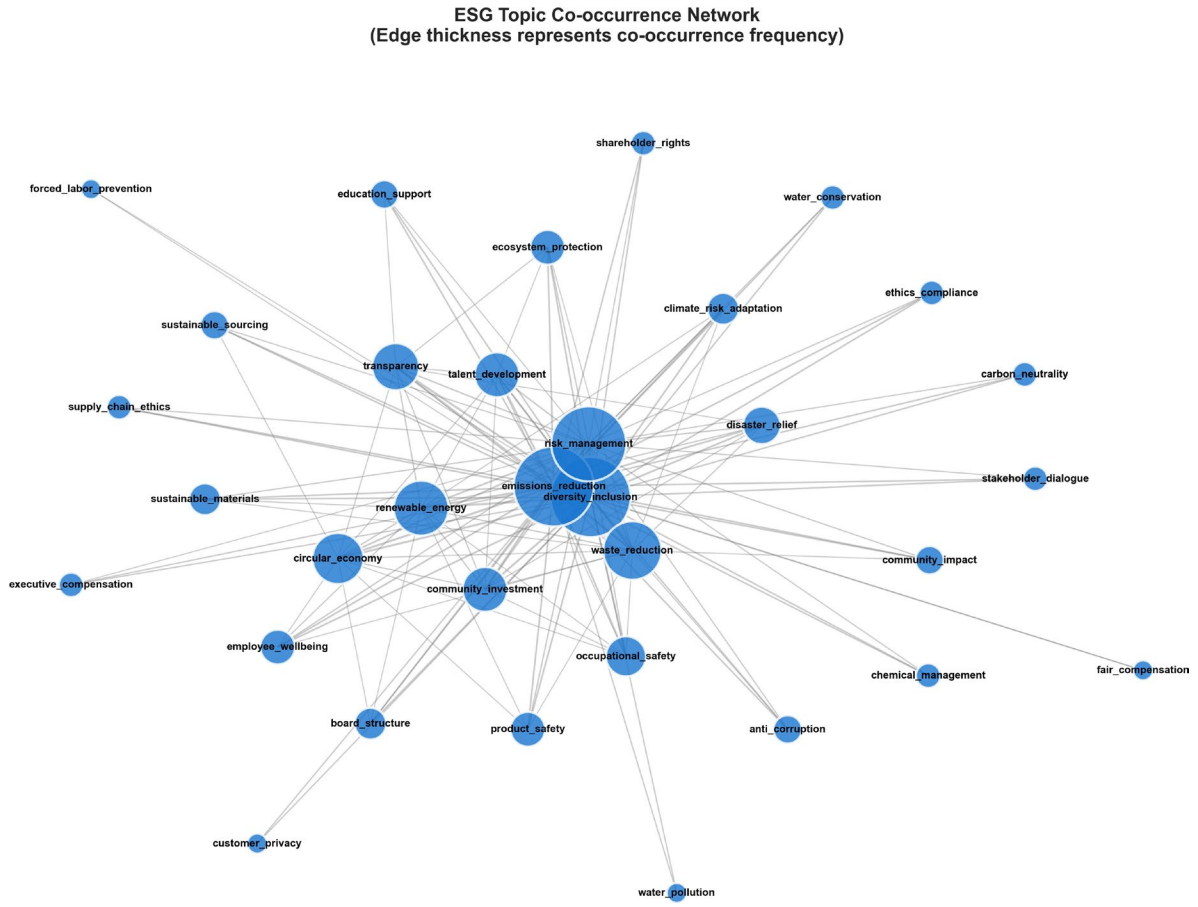


Figure 6. Topic co-occurrence network. Node size represents topic frequency; edge thickness represents co-disclosure frequency. The network reveals integrated ESG strategies with strong cross-dimensional connections, particularly between diversity and emissions reduction topics.

Table 4. Topic co-occurrence network metrics.

Metric	Value	Interpretation
Nodes (topics)	39	Complete ESG taxonomy
Edges (co-disclosures)	504	68% of possible pairs
Density	0.68	High integration
Avg. clustering coeff.	0.82	Strong triadic closure
Modularity (Louvain)	0.31	Weak community boundaries
Diameter	3	Compact network
Avg. path length	1.35	High connectivity
Assortativity	−0.12	Slight disassortative mixing

The network also exhibits strong triadic closure (average clustering coefficient = 0.82), significantly exceeding the random expectation of 0.68. Triadic closure means that when two companies both disclose topic X, they likely disclose overlapping additional topics. This pattern indicates ESG topics cluster into thematic bundles – firms select coherent portfolios rather than independent topics.

The network exhibits high connectivity, measured by two metrics. First, the network diameter (3) and average path length (1.35) indicate that any two topics are connected through at most one intermediary topic. For example, a specialized topic like 'water management' connects to 'employee wellbeing' through a single intermediary topic such as 'emissions reduction'. This compactness reflects comprehensive ESG disclosure – firms rarely report isolated single topics but disclose broad portfolios spanning multiple dimensions.

Second, the slightly negative assortativity coefficient (−0.12) reveals the network's integration pattern. Assortativity measures whether high-degree nodes (topics co-disclosed with many other topics) preferentially connect to each other. Negative assortativity indicates *disassortative mixing* – hub topics like emissions reduction (connected to 38 other topics) and diversity (connected to 37 topics) bridge

connections to both other hub topics and peripheral specialized topics like 'indigenous rights' (connected to only 12 topics). This pattern facilitates cross-domain integration: core ESG topics serve as connectors linking diverse thematic areas, rather than forming exclusive clusters of only highly popular topics.

Network centrality analysis identifies which topics serve as integration hubs by measuring two dimensions of importance. *Degree centrality* counts direct connections – how many other topics a given topic is co-disclosed with. *Betweenness centrality* measures how often a topic appears on the shortest path connecting two other topics, indicating its role as a bridge in the network.

Emissions reduction emerges as the dominant hub topic, exhibiting the highest degree centrality (connected to all 38 other topics) and betweenness centrality (67%). The betweenness score indicates that when tracing connections between any two ESG topics, emissions reduction appears on the connecting path in two-thirds of cases. For instance, the connection between 'customer privacy' (a social topic) and 'renewable energy'(an environmental topic) typically routes through emissions reduction, as firms co-disclose these topics together. This positioning makes emissions reduction the core ESG topic – nearly universally co-disclosed with all other themes.

Diversity and inclusion ranks second (degree = 37, betweenness = 61%), confirming its centrality in Japanese ESG disclosure (Kato & Kodama, 2018). In contrast, specialized topics like 'indigenous rights'(degree = 12) and 'tax transparency' (degree = 14) occupy peripheral positions, co-disclosed primarily with governance topics but rarely bridging connections to environmental or social themes.

4.2.2. Topic clustering and integration patterns

To assess whether firms' actual disclosure practices align with theoretical ESG taxonomy boundaries, we applied Louvain community detection algorithm to the topic co-occurrence network. This algorithm groups topics based on co-disclosure patterns – topics frequently disclosed together by the same companies cluster together, independent of their pre-assigned E/S/G labels in the taxonomy (Section 3.3).

Louvain identified three primary clusters, though boundaries are blurred (modularity = 0.31, where values >0.3 indicate detectable community structure but <0.5 suggest overlapping boundaries). Notably, the data-driven clusters largely align with the theoretically-grounded taxonomy. The environmental Cluster (11 topics) comprises emissions, renewable energy, water management, waste reduction, circular economy, biodiversity and climate adaptation. The social cluster (18 topics) encompasses diversity, employee wellbeing, human rights, labor practices, community engagement, product safety and customer privacy. The governance cluster (10 topics) includes board governance, risk management, compliance, anti-corruption, data security and stakeholder engagement.

The low modularity (0.31) indicates significant cross-cluster connections – firms do not rigidly separate E-S-G dimensions in practice but integrate them. This convergence between theoretical taxonomy (based on GRI, SASB and Japanese institutional literature) and empirical clustering (derived from actual disclosure co-occurrence) validates the taxonomy design while revealing that firms conceptually link topics within dimensions but operationally integrate across dimensions.

Analyzing cross-cluster edges reveals systematic integration patterns (Table 9 in Supplementary Appendix E). The most frequent topic pair is 'Diversity & Inclusion' – 'Emissions Reduction', 'co-disclosed by 95.7% of companies' (135 of 141). This E-S integration likely reflects corporate sustainability narratives linking climate action with social equity (e.g. 'just transition' framing that connects emission reductions to workforce reskilling). The second most common pairs – 'Emissions Reduction' – 'Risk Management' and 'Diversity & Inclusion' – 'Risk Management'(both 70.9%) – represent E-G and S-G integrations, positioning risk management as a governance anchor linking environmental and social actions to enterprise risk frameworks.

Environmental–social integration is particularly strong: 8 of the top 15 co-occurrence pairs span E-S boundaries, while only 3 span E-G and 4 span S-G. This pattern suggests Japanese firms conceptually link environmental and social ESG more tightly than governance, possibly reflecting stakeholder capitalism logics where environmental stewardship and employee welfare are viewed as complementary ethical obligations, whereas governance is framed as compliance infrastructure.

4.2.3. Company similarity network and strategic archetypes

To identify strategic archetypes – groups of companies with similar ESG disclosure profiles – the analysis constructed a company similarity network and applied clustering (Figure 7). Each company was

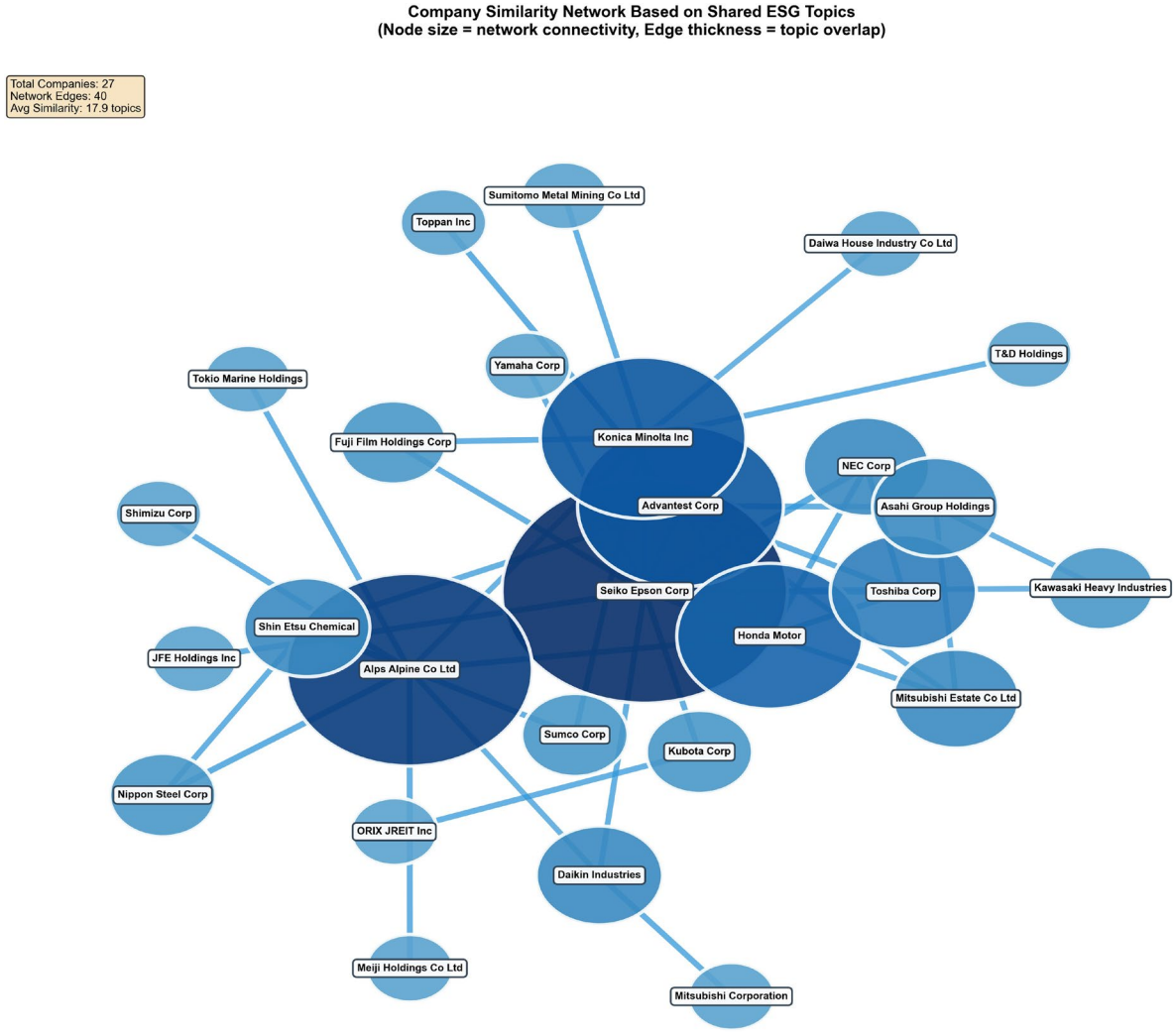


Figure 7. Company similarity network based on ESG topic profiles. Companies are positioned by similarity in disclosed topics. Three clusters emerge corresponding to balanced, social-oriented and environment-oriented archetypes.

Table 5. ESG configuration archetypes (n = 118 companies). Archetype classification was performed on 118 companies (83.7% of the 141-company sample) with sufficient action counts(n ≥ 100) for reliable clustering. Companies with fewer than 100 disclosed actions were excluded from archetype analysis to ensure statistical robustness.

Archetype	*n*(%)	E:S:G ratio	Representative industries	Example firms
Balanced	80 (67.8%)	25:34:41	Diversified industries	Itochu, Konica Minolta
Social-oriented	16 (13.6%)	16:58:26	Finance, retail, services	Mitsui Chemicals, FamilyMart
Environment-oriented	22 (18.6%)	51:23:26	Manufacturing, energy	Suntory, Daikin

represented as a 39-dimensional vector where element *i* equals the count of actions disclosed for topic *i* (see Section 3.6). Cosine similarity was computed between all company pairs, measuring ESG topic portfolio alignment (cosine similarity ranges from 0 for completely different profiles to 1 for identical profiles). A network was constructed connecting companies with similarity >0.70 (empirically chosen threshold yielding interpretable clusters while filtering weak similarities). The resulting network comprises 141 nodes with average degree 18.3, indicating companies cluster into neighborhoods of similar ESG strategies.

To formally identify clusters, the Louvain algorithm (Blondel et al., 2008) was applied – a widely-used community detection method partitioning networks by maximizing modularity (the density of connections within clusters relative to between clusters). Louvain iteratively groups nodes (companies) to maximize within- cluster ESG profile similarity. The algorithm identified three distinct archetypes (Table 5),

validated by strong industry alignment (ARI = 0.64, indicating substantial agreement between ESG-based clusters and industry sectors, well above the random baseline of ARI = 0 and approaching the ARI = 0.70–0.80 range typically considered strong agreement in clustering validation literature (Xuan Vinh et al., 2010)).

The Balanced Archetype (67.8%, n = 80) exhibits an E:S:G ratio of 25:34:41, representing relatively balanced attention across dimensions with slight governance emphasis. This largest group includes diverse industries from manufacturing to technology (Itochu, Konica Minolta, Asahi Group). These firms disclose comprehensive ESG portfolios spanning multiple dimensions, likely reflecting institutional investor pressures for holistic ESG performance and diversified business model exposure to varied ESG risks. The governance emphasis may reflect Japan's ongoing corporate governance reforms and increased board independence requirements (Kato & Kodama, 2018).

The social-oriented archetype (13.6%, n = 16) exhibits an E:S:G ratio of 16:58:26, demonstrating strong social dominance. Concentrated in finance, retail and services sectors (Mitsui Chemicals, FamilyMart, Tokio Marine Holdings), these firms prioritize employee relations, customer engagement and community investment over environmental issues. This reflects labor-intensive business models where human capital and stakeholder relationships constitute primary value drivers, and environmental footprints are relatively light (offices, branches, retail stores *versus* manufacturing facilities).

The environment-oriented archetype (18.6%, n = 22) emphasizes environmental topics with an E:S:G ratio of 51:23:26. Dominated by manufacturing, materials and energy-intensive firms (Suntory, Daikin, JFE Holdings, Nippon Steel), these companies face direct regulatory pressures (emission standards, carbon pricing), supply chain scrutiny from global clients, and physical climate risks to production facilities. Environmental disclosure intensity reflects material risk exposure from carbon-intensive operations (Khan et al., 2016). Representative ESG topic profiles for each archetype are visualized through radar charts in Figure 8 (Supplementary Appendix F). The bipartite network structure connecting companies to ESG topics is shown in Figure 9 (Supplementary Appendix F), revealing disclosure diversity and topic concentration patterns.

Archetype differences extend beyond dimensional balance to topic specificity. Environment-oriented firms are 3.2 times more likely to disclose Scope 3 emissions (42% *versus* 13% for social-oriented firms) and 2.8 times more likely to address circular economy (38% *versus* 14%), reflecting supply chain decarbonization imperatives. Social-oriented firms are 2.1 times more likely to disclose financial inclusion (31% *versus* 15% for environment-oriented) and 1.9 times more likely to address customer privacy (47% *versus* 25%), aligning with service sector business models. Complete company assignments to archetypes, including industry sectors and action counts, are provided in Table 12 (Supplementary Appendix G). The industry sector mapping used for validation is detailed in Table 13 (Supplementary Appendix G), while archetype characteristics are summarized in Table 14 (Supplementary Appendix G).

The emergence of distinct archetypes challenges universal ESG frameworks (e.g. GRI, SASB) that assume uniform topic applicability. These findings suggest ESG disclosure is contingent on business model materiality – manufacturing firms cannot credibly signal sustainability without environmental depth, while service firms require social disclosure depth. This contingency validates Khan et al. (2016) materiality thesis: financially material ESG issues vary systematically by industry, and disclosure should reflect this heterogeneity rather than imposing uniform standards.

To validate that our ESG-based archetypes reflect genuine strategic patterns rather than arbitrary clustering artifacts, we assessed whether archetype membership aligns with industry sectors. This analysis computed the ARI comparing two groupings: (1) the three ESG archetypes derived from Louvain clustering of disclosure profiles (balanced, social-oriented, environment-oriented), and (2) the 11 industry sectors from GICS/TSE classifications (Section 3.6). ARI measures clustering agreement while adjusting for chance: ARI = 0 indicates random agreement, ARI = 1 indicates perfect alignment, and ARI > 0.6 is considered substantial agreement in validation literature (Xuan Vinh et al., 2010).

This analysis yields ARI = 0.64, indicating strong industry-archetype alignment: manufacturing firms predominantly cluster into the environment-oriented archetype (81% of manufacturing firms), financial firms into the social-oriented archetype (78% of finance firms) and diversified conglomerates into the balanced archetype (72% of conglomerates). This alignment confirms that ESG disclosure archetypes are not random patterns but reflect systematic differences driven by industry-specific materiality and

iso-morphic pressures within sectors (DiMaggio & Powell, 1983). Firms in the same industry face similar regulatory environments, stakeholder expectations, and operational risks, leading to convergent ESG disclosure strategies.

4.3. Validation and robustness

Manual validation of 100 stratified random samples yielded 89.5% average accuracy (all eight record elements correct; Cohen's $\kappa = 0.95$ between two independent coders, indicating almost perfect agreement). Accuracy increased monotonically with confidence: records with confidence = 1.0 showed 97.7% accu- racy, 0.70–0.79 showed 85.7% and 0.60–0.69 showed 77.5%, validating our 0.60 threshold effectively separates reliable from unreliable extractions. Error analysis revealed four primary failure modes:(1) *overly abstract actions* – action verbs lacking specific objects(e.g. 'implement' without stating what is implemented), accounting for 54.5% of errors;(2) *incorrect target identification* – misidentifying action beneficiaries(18.2%);(3) *compound actions* – single textual statements describing multiple distinct actions that should have been extracted as separate records(18.2%); (4) *evidence text mismatches* – extracted text not appearing verbatim in source documents(9.1%).

Topic-level confidence varies (Table 11 in Supplementary Appendix E): Governance topics average 0.9984 (highest), social 0.9951 and environment 0.9925 (lowest). This may reflect governance terminology standardization (e.g. 'independent director', 'audit committee') *versus* environmental jargon diversity(multiple terms for similar concepts).

5. Discussion

5.1. Theoretical interpretation

These findings engage with three theoretical perspectives central to ESG disclosure research.

Stakeholder Salience Theory (Freeman, 1984; Mitchell et al., 1997) predicts that firms prioritize stakeholders who possess three attributes: *power*(ability to influence firm outcomes), *legitimacy*(socially accepted claims on the firm) and *urgency* (time-sensitive demands). Prior studies infer stakeholder prioritization from survey data (Gond et al., 2011) or binary disclosure indicators (Dhaliwal et al., 2011). The actor–target extraction enables direct observation of which stakeholders receive disclosed attention, operationalizing salience through action frequency.

Signaling Theory (Connelly et al., 2011) distinguishes credible commitment signals from symbolic rhetoric. Credible signals are costly and verifiable (e.g. 'reduced emissions by 20% through $50 M facility upgrade'), while symbolic claims are low-cost and unverifiable (e.g. 'committed to sustainability'). Prior research tests signal credibility by examining market reactions (Ioannou & Serafeim, 2017) or performance correlations (Cao et al., 2023). The modality extraction (past/ongoing/future) provides a complementary lens: completed actions may signal greater commitment than future plans, as they are costlier to fabricate.

Institutional Isomorphism (DiMaggio & Powell, 1983) posits that organizations facing similar regulatory and normative pressures adopt similar practices, resulting in industry-level convergence. Prior studies document such convergence through disclosure topic frequencies (Christensen et al., 2022; Lin et al., 2024). Company similarity networks test whether disclosure portfolios cluster by industry, revealing the extent to which ESG reporting reflects sectoral norms.

These theoretical perspectives have traditionally been tested using aggregated measures such as ESG ratings, binary disclosure indicators or topic frequency counts. Such coarse-grained data limit the ability to validate the specific mechanisms proposed by each theory. Action-level knowledge graphs enable more direct tests. Extracting *who performs actions targeting whom* tests stakeholder salience theory's prioritization predictions. Capturing *whether actions are described as completed or planned* assesses signaling theory's credibility dynamics. Examining *whether companies in the same industry exhibit similar disclosure portfolios* evaluates institutional isomorphism. These findings both support and qualify these theoretical predictions, as detailed below.

5.1.1. Stakeholder salience and prioritization hierarchies

Stakeholder theory (Freeman, 1984; Mitchell et al., 1997) predicts firms prioritize stakeholders exhibiting power, legitimacy, and urgency. The actor–target extraction operationalizes this theory with unprecedented granularity. The employee-centric pattern(10.2% of actions target employees *versus* 0.5% local communities, 0.3% customers) confirms employees as 'definitive stakeholders' possessing all three salience attributes in Japanese firms: power(collective bargaining, productivity control), legitimacy (legally recognized employment relationships) and urgency (labor shortages amplify retention pressures) (Kato & Kodama, 2018).

This employee dominance aligns with Japan's stakeholder capitalism tradition (Buchanan & Deakin, 2024; Miyajima & Saito, 2021), where firms historically prioritize stable employment over shareholder returns. However, these findings reveal *selective stakeholder disclosure*: while firms extensively report employee-focused actions (10.2% of all targets), external stakeholder engagement receives limited disclosure. Communities (0.5%), NGOs (0.8%) and indigenous groups(negligible) appear infrequently as action targets despite possessing legitimacy and urgency attributes according to stakeholder theory (Mitchell et al., 1997).

This pattern suggests two non-mutually exclusive explanations. First, *disclosure selectivity*: firms may genuinely engage external stakeholders through less-reported channels(informal community dialogues, NGO partnerships not formalized in reports) while over-reporting formalized employee programs. Second, *hierarchical stakeholder salience*: proximate, economically integrated stakeholders (employees, suppliers at 10.2% and 1.5%) may receive genuine priority over distant civil society actors, consistent with stakeholder salience theory's prediction that proximity and economic interdependence amplify attention (Gond et al., 2011; Mitchell et al., 1997). The data cannot distinguish whether this reflects disclosure bias or genuine resource allocation patterns – triangulation with external data (e.g. community investment expenditures, NGO partnership agreements) would be required to separate disclosure from action.

Notably, despite Japan's advanced age crisis (Kato & Kodama, 2018), 'senior citizens' or 'elderly' appear in only 1.2% of actions, suggesting firms respond to labor supply challenges more through workforce retention policies than elder-focused social programs. Similarly, 'communities' and 'NGOs' receive relatively low attention (3.1% and 0.8%, respectively), indicating external stakeholder engagement remains underdeveloped compared to internal (employee) focus.

The supplier-as-target pattern (1.5%) reveals asymmetric power dynamics. Firms disclose actions *toward* suppliers (audits, training, guideline dissemination) rather than actions *by* suppliers, positioning supply chain ESG as hierarchical governance rather than collaborative partnership (Table 10 in Supplementary Appendix E details the most common actor–target combinations). This contradicts stakeholder partnership rhetoric suppliers appear as compliance targets, not co-creators of sustainability value.

5.1.2. Signaling theory and credibility dynamics

Signaling theory (Connelly et al., 2011) distinguishes credible signals (costly, hard to mimic) from cheap talk(low-cost claims easily faked). Prior ESG disclosure research tests signal credibility by examining whether disclosure correlates with actual performance (Cao et al., 2023) or market reactions (Ioannou & Serafeim, 2017), interpreting strong correlations as evidence of credibility. The modality distribution enables a complementary approach: examining temporal framing patterns as indicators of rhetorical commitment.

The dominance of 'ongoing' actions (65.5%) over 'future commitments' (19.1%) challenges common greenwashing critiques that emphasize aspirational pledges without implementation (Lyon & Maxwell, 2011). Japanese firms predominantly disclose operational activities rather than forward-looking plans, which may suggest relatively higher signal credibility compared to commitment-heavy disclosure profiles. However, this pattern may also reflect reporting framework composition and scope boundaries. Japanese sustainability reports predominantly follow GRI Standards, which emphasize demonstrated actions and achieved performance outcomes rather than forward-looking projections (Benameur et al., 2024). In contrast, forward- looking climate commitments are often disclosed in separate TCFD-aligned climate reports or integrated reports (TCFD Consortium, 2022), which were not included in our analytical scope. This methodological boundary suggests interpreting modality distributions requires triangulation across multiple report types to capture comprehensive disclosure patterns (de Villiers et al., 2014).

However, the low proportion of 'past_action' completed outcomes (15.4%) relative to 'ongoing' activities (65.5%) warrants explanation. Three complementary factors may account for this pattern.

First, *framework-driven temporal framing*: GRI Standards' emphasis on quantitative performance indicators (e.g. GRI 305 Emissions, GRI 401 Employment) structurally favors reporting of current-state metrics and ongoing operational activities over aspirational future targets (Benameur et al., 2024). Actions coded as 'ongoing' often represent continuous operational processes (e.g. 'implementing energy efficiency measures', 'providing employee training') that align with GRI's indicator-based reporting structure, whereas strategic multi-year initiatives may be aggregated at higher levels in integrated reports following IIRC frame-works (Eccles & Krzus, 2014).

Second, *reporting granularity differences*: Complex, multi-year ESG transitions(e.g. carbon neutrality pathways, diversity transformations) are often reported at strategic level in integrated reports or management discussion sections, whereas sustainability reports emphasize operational-level actions amenable to 'ongoing' classification (de Villiers et al., 2014). This reporting structure reflects different stakeholder audiences – sustainability reports target ESG specialists seeking operational metrics, while integrated reports address investors seeking strategic direction (Eccles & Krzus, 2014).

Third, *reporting recency bias*: Sustainability reports typically emphasize current-year activities and forward commitments to demonstrate ongoing engagement with ESG issues. Historical achievements completed in prior years may be mentioned briefly in retrospective summaries but receive less detailed action-level documentation, leading to their underrepresentation in the extracted dataset.

The substantially higher 'future_commitment' ratio in environmental disclosure (28.3%) compared to social (16.1%) or governance (12.8%) topics reflects climate transition's long time horizons and intensifying regulatory pressures. Japanese firms face mandatory TCFD disclosure requirements (Government of Japan, 2021) and Paris Agreement decarbonization commitments, which structurally incentivize forward-looking climate pledges spanning decades (e.g. net-zero by 2050). However, this analysis scope – sustainability reports only – may underestimate total climate commitment disclosure, as many Japanese firms publish standalone TCFD reports or disclose scenario analyses in integrated reports (TCFD Consortium, 2022). Future research analyzing multiple report types concurrently could provide more comprehensive assessment of forward-looking climate disclosure patterns. This temporal structure differs from social or governance initiatives that often have shorter implementation cycles.

However, commitment-heavy disclosure without corresponding 'ongoing' or 'past_action' evidence may indicate symbolic compliance rather than substantive implementation – a pattern consistent with greenwashing literature's distinction between symbolic adoption and operational integration (Lyon et al., 2024). Conversely, action-heavy disclosure with limited future commitments – as observed in this Japanese sample – may reflect either genuine operational focus or methodological artifacts from analyzing GRI-structured sustainability reports that prioritize current-state performance (Christensen et al., 2022). Distinguishing these interpretations requires triangulating sustainability reports with forward-looking disclosures in other report types. Recent research on mandatory climate disclosure shows that firms respond to compliance pressures by expanding forward-looking statements while implementation lags (Lin et al., 2024), suggesting disclosure may precede action by multiple years.

This framework enables stakeholders to construct temporal consistency metrics, such as the ratio of completed actions to forward commitments within each topic domain. Firms exhibiting systematically high commitment-to-action ratios (e.g. numerous 2030 targets but few documented implementation steps) merit closer scrutiny regarding implementation credibility. This approach operationalizes signaling theory's core insight: credible signals require observable costs or constraints that distinguish genuine commitment from cheap talk.

Illustrative Case: Temporal Inconsistency in Climate Disclosure. To demonstrate the analytical value of action-level granularity, consider a representative case from our dataset. One manufacturing firm[1] in our sample disclosed 107 climate-related actions across emissions reduction, renewable energy, and energy efficiency topics. However, modality analysis exposed substantial temporal asymmetry: 67 actions (62.6%) were classified as 'future_commitment' (e.g. long-term decarbonization targets, renewable energy investment plans), 32 actions (29.9%) as 'ongoing' (e.g. implementing energy management systems), and only 8 actions (7.5%) as 'past_action' with quantified outcomes(e.g. achieved emission reductions with baseline comparisons). This 8.4:1 commitment-to-completion ratio substantially exceeded the sample median

of 1.9:1, suggesting disclosure emphasis on aspirational pledges rather than demonstrated implementation. Moreover, the eight completed actions predominantly referenced internal metrics without third-party verification, while the 67 future commitments varied widely in specificity – some specified interim milestones and verification mechanisms, others stated general directional goals without accountability frameworks. In contrast, peer manufacturing firms at the sample median exhibited more balanced modality distributions(approximately 45% ongoing, 37% future, 19% past) based on our full-sample analysis. This case illustrates how action-level analysis can surface temporal consistency patterns – the balance between commitment disclosure and implementation evidence – that aggregate ESG ratings may not capture. It does not constitute a judgment of greenwashing (which would require triangulating disclosure with external performance data such as verified emissions registries), but rather demonstrates how stakeholders can systematically assess disclosure credibility through modality distribution analysis as a complement to overall ratings.

An important limitation concerns the distinction between disclosure quantity and action quality. This method counts the number of disclosed actions but cannot assess their substantive impact. For example, a firm reporting 100 brief employee training sessions may represent lower ESG investment than a firm reporting 10 intensive multi-week programs, yet both register as multiple 'train' actions in the dataset. Similarly, a firm disclosing 'reduced emissions' achieves the same extraction as a firm specifying 'reduced Scope 1-2-3 emissions by 42% with third-party verification' – despite vastly different credibility levels.

This measurement challenge reflects a fundamental trade-off in automated ESG analysis. Granularity in action counting enables scalability (analyzing 77,734 records across 141 firms), but granularity in action evaluation requires context-specific human judgment that does not scale. Addressing this limitation requires triangulation: the knowledge graph identifies *what firms claim* (disclosure patterns), while external data sources – carbon registries, labor statistics, regulatory filings – can validate *what firms do*(actual performance). Integrating disclosure-based knowledge graphs with performance-based verification represents a critical direction for future research.

5.1.3. Institutional isomorphism and disclosure conformity

The emergence of three distinct ESG archetypes (balanced, social-oriented, environment-oriented) with strong industry alignment (ARI = 0.64) among Japanese firms validates institutional theory's (DiMaggio & Powell, 1983) prediction of isomorphic pressures. Institutional isomorphism theory posits that organizations facing similar environmental conditions adopt similar structures and practices to gain legitimacy (DiMaggio & Powell, 1983), with prior ESG research documenting industry-level convergence in disclosure topics (Christensen et al., 2022) and reporting frameworks (Lin et al., 2024). This Japanese context provides particularly strong evidence for isomorphism, as Japan's coordinated market economy features dense inter-firm networks, industry associations and long-term business relationships that facilitate normative consensus and mimetic diffusion (Buchanan & Deakin, 2024).

These findings reveal industry-specific isomorphic mechanisms in Japanese ESG disclosure. Manufacturing firms converge on environment-oriented disclosure through coercive isomorphism and mimetic isomorphism. Mandatory compliance with emission regulations, carbon pricing schemes, and TCFD requirements (Government of Japan, 2021) compels environmental reporting as a coercive mechanism. Simultaneously, firms facing strategic uncertainty emulate visible industry leaders' climate disclosure practices to signal legitimacy through mimetic processes. Financial firms converge on social-oriented disclosure through normative isomorphism: professional communities(banking associations, asset management networks) propagate shared norms emphasizing human capital, customer data governance, and stakeholder trust as core value drivers (Ioannou & Serafeim, 2017), leading to coordinated emphasis on social topics.

However, the *low modularity* (0.31) of the topic co-occurrence network reveals that isomorphism operates at the intensity level (E-S-G emphasis) rather than the selection level (topic presence/absence). Nearly all firms (>95%) disclose the same core topics (emissions reduction, workforce diversity, board independence), but differ substantially in action counts per topic. This convergence in topic selection but divergence in topic emphasis suggests that institutional pressures have established a *mandatory disclosure template* – a bounded set of topics firms must address to maintain legitimacy – while allowing flexibility in how extensively each topic is covered.

This pattern reflects institutional standardization's dual effects documented in organizational sociology (DiMaggio & Powell, 1983). First, *enhanced comparability*: uniform topic coverage enables investors to benchmark firms on common dimensions (e.g. all firms report climate actions, facilitating emission intensity comparisons). However, second, *constrained differentiation*: firms may face difficulty signaling unique ESG capabilities if they disclose the same topic portfolio as competitors. Organizations operating in highly institutionalized fields often sacrifice strategic distinctiveness for legitimacy (DiMaggio & Powell, 1983), potentially leading to homogeneous disclosure that obscures genuine performance differences.

The strong E-S integration (8 of top 15 co-occurrence pairs span Environmental-Social boundaries) challenges traditional three-pillar ESG frameworks that treat E-S-G as independent dimensions. Firms increasingly link environmental and social issues through integrated narratives: 'just transition' couples climate action with labor reskilling, 'green jobs' frames renewable energy as employment creation, and 'environmental justice' connects pollution reduction to community health. This cross-domain linkage reflects two developments in corporate sustainability discourse.

First, regulatory frameworks increasingly mandate integrated thinking. The EU Corporate Sustainability Reporting Directive (CSRD) requires firms to analyze 'double materiality' – how ESG issues affect both financial performance *and* stakeholders (Lin et al., 2024) – naturally leading to cross-domain connections (e.g. emissions affect both compliance costs and community health). Second, firms facing stake- holder pressure on multiple fronts may strategically bundle ESG actions to satisfy diverse audiences: framing renewable energy investment as simultaneously climate mitigation (environmental) and job creation (social) appeals to both environmental NGOs and labor unions.

However, integration complicates categorical analysis. Should 'supplier diversity training' be classified as Social(diversity), social (supplier relations) or both? Should 'electric vehicle fleet adoption' count as environmental (emissions) or social (air quality impacts on communities)? Traditional rating methodologies resolve this through forced categorization, potentially losing information. Our knowledge graph design preserves integration nuances through multiple topic tagging and network relationships, enabling users to query cross-domain patterns (e.g. 'find all actions simultaneously tagged Environmental and Social') rather than imposing mutually exclusive classifications.

5.2. Methodological implications

This LLM-driven knowledge graph approach addresses longstanding limitations of ESG disclosure measurement while introducing new considerations for future research.

5.2.1. Advantages over traditional approaches

Compared to rule-based extraction, LLMs excel at semantic understanding without requiring exhaustive pattern libraries. Traditional regex or dependency parsing approaches struggle with linguistic variation: 'reduced emissions by 20%', 'achieved 20% emission reduction' and 'cut carbon footprint one-fifth' express semantically identical actions but require distinct extraction rules. LLMs recognize semantic equivalence through contextual embeddings, substantially reducing manual rule engineering.

Compared to supervised machine learning (Named Entity Recognition and Relation Extraction models), our zero-shot approach eliminates the need for large labeled training datasets – critical for ESG where expert annotation is costly and domain-specific datasets remain scarce. GPT-4o's zero-shot instruction following achieved 89.5% accuracy (Cohen's $\kappa = 0.95$), approaching performance of supervised NER models trained on thousands of annotated documents while requiring only prompt engineering rather than model fine-tuning.

However, LLMs introduce a transparency challenge analogous to ESG ratings' opacity. While ESG ratings obscure measurement processes (e.g. how MSCI weights different metrics), LLMs obscure reasoning processes. When GPT-4o infers 'employees' as the actor from the phrase 'our workforce', the internal token representations and attention patterns driving this inference remain inaccessible to researchers. The current mitigation– requiring evidence citation for each extracted record – enables *post-hoc* verification (human coders can check whether 'our workforce' justifies 'employees') but does not provide real-time interpretability.

Addressing this limitation, emerging explainable AI techniques offer promising directions. Attention visualization can identify which input tokens GPT-4o weighs most heavily when extracting specific elements (Pan et al., 2024), while chain-of-thought prompting can require models to articulate intermediate reasoning steps before finalizing extractions. Future implementations might combine structured output with reasoning traces, generating outputs like: 'Actor: employees (reasoning: 'our workforce' refers to employee collective; evidence: page 42, line 3)'. Such approaches could enhance both verifiability and methodological transparency.

5.2.2. Knowledge graph advantages and limitations

Knowledge graphs offer superior flexibility over tabular data for ESG analysis. In relational databases, representing multi-topic actions (e.g. 'supplier diversity training' touching both diversity and supply chain topics) requires junction tables and complex joins. Knowledge graphs natively support multiple topic tags via edges, enabling natural representation of ESG integration. Network queries ('find all actions connecting employees to environmental topics') are straightforward graph traversals but cumbersome SQL.

However, knowledge graphs incur computational costs. The 77,734-record graph with 233,000 relationships fits in-memory (Neo4j), but scaling to multi-year, multi-country analysis(millions of records) may require distributed graph databases or hybrid storage strategies. Additionally, knowledge graphs privilege relationship analysis over statistical inference – regression models or causal inference are more naturally conducted on tabular data exported from graph queries. Optimal ESG research infrastructure likely combines knowledge graphs (for exploratory pattern discovery) with data warehouses (for confirmatory statistical testing), leveraging each representation's strengths.

5.2.3. Cross-jurisdictional generalizability: universal components versus context-specific adaptation

While this empirical application focuses on Japanese firms, the methodological framework exhibits distinct layers of generalizability. The following analysis clarifies which components transfer universally *versus* which require jurisdiction-specific adaptation to broaden the approach's international applicability.

5.2.3.1. Universally applicable components. Three core methodological elements transfer across all jurisdictions without modification. The LLM-based extraction pipeline employs GPT-4o and similar multilingual LLMs (Claude, Gemini, Qwen) that process sustainability narratives in English, Japanese, Chinese, Spanish and dozens of other languages uniformly. The zero-shot instruction-following paradigm remains constant; only the input language changes. The action-level schema – the eight-element structure (company, topic, actor, action, target, modality, evidence and confidence) – represents universal semantic categories applicable to any sustainability disclosure, independent of reporting framework or cultural context. Whether analyzing EU CSRD double-materiality reports or U.S. SEC climate risk disclosures, the fundamental question 'who does what, for whom, when, and with what evidence'? remains analytically relevant. The knowledge graph infrastructure provides graph-based representation (nodes = entities, edges = relationships) and network analysis methods (centrality, community detection, path queries) that are mathematically universal, applicable regardless of disclosure content or jurisdiction.

5.2.3.2. Japan-specific elements requiring cross-jurisdictional adaptation. Three aspects of this implementation reflect Japanese disclosure norms and require modification for other contexts. Bilingual report availability characterizes Japanese Prime Market firms, which frequently publish bilingual sustainability reports (Japanese + English), enabling cross-language validation that strengthened extraction accuracy. Jurisdictions with predominantly monolingual disclosure (e.g. China, France, Brazil) require either native-language LLMs (Qwen for Chinese, Mistral for French) or translation preprocessing, introducing potential semantic drift. Process-oriented disclosure culture reflects Japanese firms' emphasis on continuous improvement narratives (kaizen philosophy), which may inflate 'ongoing' modality proportions (65.5% in these data) relative to Western firms favoring milestone-based disclosure. Cultural framing differences affect modality distributions but do not invalidate the analytical framework – modality analysis remains interpretable, though baseline expectations may differ. Reporting framework composition represents a third adaptation requirement: the 39-topic taxonomy (Supplementary Appendix B) blends GRI(dominant in Japan), SASB(sector-specific materiality), and TCFD (climate risk). Jurisdictions with

different regulatory regimes require taxonomy customization, as detailed below.

5.2.3.3. Cross-jurisdictional deployment strategy: three-step adaptation. Deploying this methodology beyond Japan requires systematic localization through three interconnected steps.

Taxonomy Localization modifies the 39-topic list to reflect jurisdiction-specific materiality frameworks and regulatory requirements. For EU CSRD applications, adding double-materiality topics emphasizing impact assessment (e.g. biodiversity footprint, social taxonomy alignment) alongside financial materiality becomes essential. For U.S. SEC climate disclosure rules, the approach emphasizes financially material climate risks and Scope 1–3 emissions quantification while de-emphasizing broad stakeholder engagement topics. For Chinese CSR reports, integration of national priorities (e.g. common prosperity initiatives, carbon neutrality pathways aligned with 2060 targets, Belt and Road sustainability) ensures contextual relevance. Critically, taxonomy modification requires *only prompt engineering* – updating the topic list in extraction instructions – not model retraining, preserving the zero-shot advantage.

Language Model Selection deploys native-language LLMs for monolingual disclosures to preserve cultural and linguistic nuances. Chinese sustainability reports benefit from models like Qwen or GLM- 4 trained on Chinese corporate discourse, capturing concepts like 'UTF8gbsn' (ecological civilization) or 'UTF8gbsn' (harmonious society) that English-centric LLMs may misinterpret. Similarly, Arabic ESG reports (increasingly common in Gulf Cooperation Council countries) require Arabic-capable models to handle concepts like Islamic finance principles or waqf (endowment) sustainability. For bilingual or English-language reports (common in multinational corporations and international financial centers), GPT-4o, Claude or Gemini remain effective.

Validation Calibration re-validates extraction accuracy using jurisdiction-specific expert annotations, as disclosure rhetoric varies by reporting culture. For example, German firms' technical precision in environmental reporting may yield higher evidence specificity than narrative-heavy U.S. CSR reports, affecting confidence score distributions. Establishing jurisdiction-specific validation benchmarks (analogous to the 89.5% accuracy for Japanese reports) ensures methodological reliability across contexts.

5.2.3.4. Disclosure maturity and report structure constraints. This approach assumes structured, comprehensive sustainability reports averaging 68 pages (range: 22–156 in the sample) with clearly delineated ESG sections. This assumption holds for large listed firms in developed markets (North America, Western Europe, Japan, Australia) where sustainability reporting is mature or mandatory. However, firms in emerging markets or small-to-medium enterprises may produce sparse disclosures (e.g. 10–20 page PDFs with scattered sustainability mentions integrated into annual reports), unsuitable for automated action-level extraction at scale. For such contexts, hybrid workflows – automated extraction for structured sections combined with manual coding for unstructured narratives – represent a pragmatic compromise between scalability and coverage.

Importantly, the core methodological contribution—action-level structured extraction via zero- shot LLM prompting enabling stakeholder relationship mapping, modality analysis, and network- based integration assessment—transfers universally. Jurisdiction-specific customization affects taxonomy content (which topics matter), language processing (which LLM to use), and validation benchmarks (what accuracy threshold is acceptable), but does not alter the fundamental analytical paradigm. This universality-with-localization balance positions the framework as adaptable infrastructure for global ESG disclosure research, not a Japan-specific methodological artifact.

5.2.4. Temporal and longitudinal considerations

Single-year analysis, while providing rich cross-sectional insights, cannot address temporal questions central to ESG credibility: Do firms follow through on commitments? Are disclosure shifts driven by genuine strategic change or regulatory compliance? Distinguishing these dynamics requires multi-year knowledge graphs tracking the same firms over time. Multi-year knowledge graphs would enable three critical longitudinal analyses. Commitment tracking examines whether future_commitment actions in year T materialize as past_action in T+1, with systematic non-fulfillment patterns flagging potential greenwashing. Topic evolution analysis investigates whether firms shift disclosure emphasis over time in response to regulatory changes (e.g. TCFD adoption), stakeholder pressures, or industry trends, where time-series

clustering can identify leading *versus* lagging adopters. Causal impact evaluation assesses whether specific events (e.g. climate scandals, board changes, activist campaigns) altered disclosure patterns, with difference-in-differences or synthetic control methods applied to knowledge graph metrics estimating causal effects.

Yet longitudinal analysis faces methodological challenges: topic taxonomy changes (new ESG issues emerge), LLM model evolution (GPT-4o today *versus* GPT-5 tomorrow may extract differently), and report format shifts (integrated reports gaining popularity over standalone sustainability reports). Maintaining methodological consistency across years while adapting to evolving ESG discourse requires careful versioning and bridging strategies.

5.2.5. Validation and reliability

The 89.5% extraction accuracy (Cohen's $\kappa = 0.95$) is high but imperfect. The 10.5% error rate compounds in downstream analyses – network centrality rankings may shift, archetype assignments may change. Sensitivity analysis testing whether key findings hold when excluding low-confidence records (e.g. rerunning analysis with confidence >0.95) can assess robustness. Alternatively, ensemble approaches – running extraction with multiple LLMs (GPT-4, Claude, Gemini) and reconciling outputs through majority voting or weighted averaging – could improve reliability, though at 3–5x computational cost.

Importantly, these validation measures assess extraction accuracy (does extracted record match source text?) but not disclosure veracity (does source text reflect reality?). A firm may accurately disclose 'achieved 30% emission reduction' while cherry-picking favorable metrics or ignoring Scope 3. Knowledge graphs capture *what firms say*, not *what firms do*. Triangulation with external data (e.g. carbon registries, regulatory filings, third-party audits) is essential for assessing disclosure credibility *versus* mere disclosure patterns.

5.3. Practical implications

5.3.1. Primary user groups and decision contexts

This methodology targets two primary user groups with distinct decision-making needs: institutional investors seeking to assess ESG disclosure credibility, reputation risk and substantive commitment beyond aggregated ratings, and corporate managers aiming to benchmark disclosure practices, identify gaps relative to peers, and optimize disclosure strategy for stakeholder communication. The following sections detail specific application workflows for each group, followed by implications for regulatory policy development.

5.3.2. For corporate managers: action-level benchmarking and disclosure strategy optimization

Traditional ESG ratings provide firms with numerical scores but offer limited guidance on where deficiencies lie or how to improve. This knowledge graph enables granular benchmarking. A company can query: 'Among industry peers, what proportion of diversity actions target senior leadership *versus* general employees'? or 'How do top-rated firms balance past actions *versus* future commitments in climate disclosure'? Such queries reveal specific disclosure gaps that management can address at the operational level. For example, if analysis shows peer firms average 12% of ESG actions as supplier-focused, while a focal firm reports only 3%, management can investigate whether this reflects genuine supply chain governance gaps or merely disclosure incompleteness. This comparative analysis reveals disclosure patterns but does not prescribe optimal ratios – materiality varies by industry context, stakeholder expectations, and strategic priorities. Managers can use these patterns to identify disclosure gaps (e.g. if all peers disclose Scope 3 emissions but the focal firm does not), assess disclosure credibility (e.g. high future commitments without ongoing actions may signal greenwashing risk), and benchmark disclosure practices (e.g. comparing disclosure depth and specificity).

Critically, disclosure alignment with peers does not guarantee ESG performance adequacy. For evaluative benchmarking, managers should integrate action-level data with external standards (e.g. science-based targets initiative for emissions, industry-specific SASB metrics for materiality assessment).

Modality analysis offers strategic signaling insights. The finding that 65.5% of disclosed actions are ongoing (*versus* 19.1% future commitments) suggests that credible ESG disclosure emphasizes operational reality over aspirational pledges. Firms relying heavily on 'will implement' or 'committed to' language without corresponding 'has implemented' actions may be perceived as greenwashing. Managers can audit their disclosure modality distribution, ensuring sufficient concrete action reporting to substantiate strategic commitments. This is particularly critical as regulatory scrutiny intensifies: both the U.S. SEC and EU CSRD regulations explicitly target unsubstantiated claims.

The employee-centric disclosure pattern (10.2% of targets are employees *versus* 0.5% local communities) reflects either genuine stakeholder prioritization or disclosure bias. Firms pursuing multi- stakeholder engagement strategies should consider whether external stakeholder actions (community engagement, NGO partnerships, customer privacy protections) are under-reported. Conversely, if employee focus reflects deliberate strategy responding to labor market tightness, these findings validate this approach as industry-normative in Japan.

5.3.2.1. Operational workflow for disclosure gap analysis. Managers can systematically benchmark their ESG disclosure against peers through a structured workflow. First, query peer firms' action distribution by topic. Second, calculate coverage ratio for each topic as (focal_firm_actions/peer_median). Third, prioritize under-disclosed topics where ratio < 0.7. Fourth, cross-reference with modality distributions – if peer firms show 65.5% ongoing actions but the focal firm shows > 80% future commitments in a given topic, this warrants investigation for potential greenwashing risk. For example, the data reveals emissions reduction (15.2% of all actions, Figure 5) and diversity and inclusion(12.6%) as the two most frequently disclosed topics. A firm disclosing <5% in either category faces significant disclosure gap relative to industry norms and should prioritize enhancing disclosure in these material areas.

For Institutional Investors: Due Diligence, Reputation Risk Assessment, and Greenwashing Detection. *Decision Context:* ESG investors face two critical decisions: portfolio construction – which firms demonstrate substantive ESG commitment beyond high ratings – and risk monitoring – identifying reputation risks from disclosure-reality gaps before they materialize into controversies or regulatory penalties. Traditional third-party ratings are opaque, divergent, and slow to update, providing limited insight into either question. This knowledge graph enables investor-customizable analysis that complements ratings with disclosure-level transparency.

Application Workflow: Investors can define materiality based on their own criteria rather than accepting providers' predetermined weights. For example, a climate-focused fund can filter for emission reduction, renewable energy, and Scope 3 disclosure actions, ignoring governance topics irrelevant to climate objectives – a customization impossible with fixed-weight ESG scores.

Action-level granularity enables greenwashing detection through internal consistency checks. For instance, if a firm discloses 50 actions on 'renewable energy investment' but only 2 actions on 'emission reduction achieved', this inconsistency warrants scrutiny. Process-oriented actions (investments, committee formations) without outcome-oriented actions (actual reductions, audits) may signal symbolic rather than substantive commitment. Similarly, firms with high future_commitment ratios (e.g. > 40%) relative to peers merit investigation: are these firms undergoing genuine transitions, or systematically over-committing relative to implementation capacity?

Temporal analysis (when extended to multi-year knowledge graphs in future work) can track commitment fulfillment. Did actions coded as 'future_commitment' in 2023 materialize as 'past_action' in 2024? Systematic non-fulfillment flags credibility concerns. Investors can also leverage actor–target pat- terns: firms where 'board' actors dominate may exhibit top-down ESG governance, whereas 'employee' – actor dominance suggests grassroots engagement. Neither approach is necessarily superior, but the pattern should align with the firm's stated ESG strategy.

Greenwashing Detection Checklist. Investors can flag potential greenwashing using quantitative thresholds derived from empirical distributions in this analysis. Future commitment ratio exceeding 40% (industry median: 19.1%, Figure 3), especially in environmental topics where forward-looking claims concentrate (28.3% future commitments *versus* 16.1% social, 12.8% governance), signals disproportionate reliance on aspirational pledges over operational actions. Employee-target ratio below 5% (industry norm: 10.2%) indicates potential under-investment in human capital despite Japan's stakeholder capitalism tradition (Buchanan & Deakin, 2024; Freeman, 1984). Supply chain actions below 1% (industry: 2.8%) suggests

limited supply chain governance (Nguyen & Zuidwijk, 2025), particularly concerning for sectors with significant Scope 3 emissions or labor-intensive supply chains. It is important to note that these quantitative thresholds are illustrative benchmarks calibrated to a specific empirical context: a single fiscal year, a single national market (Japan), and a single extraction model (GPT-4o). Empirical distributions underlying these thresholds – the 19.1% median future commitment ratio, the 10.2% employee-target norm, and the 2.8% supply chain benchmark – reflect structural features of Japanese ESG disclosure rather than universal industry standards. Practitioners seeking to apply these thresholds operationally should first recalibrate them against datasets spanning multiple years, jurisdictions, and reporting frameworks.

5.3.3. For regulators: policy evaluation and standard development

Regulators designing disclosure mandates face trade-offs: prescriptive requirements enhance comparability but reduce flexibility; principles-based guidance preserves flexibility but sacrifices consistency. The methodology offers empirical evidence to inform these design choices. For example, Japan's TCFD adoption correlates with high climate disclosure frequency (21.5% of all actions), validating topic-specific mandates' effective-ness. However, the social-dominance pattern(55.4%) suggests climate-only mandates miss the disclosure majority – integrated E-S-G frameworks appear more appropriate for comprehensive disclosure coverage.

Modality standardization represents a critical policy gap. Analysis found substantial manual coding required to categorize 'committed to', 'plans to', is implementing' and 'has completed' due to linguistic ambiguity. Regulatory guidance specifying standardized modality language (akin to accounting standards' precise terminology) would enhance comparability and reduce greenwashing opportunities. For instance, requiring firms to separately tabulate (1) actions completed in reporting period, (2) ongoing multi-year initiatives and (3) forward-looking commitments would operationalize the framework's credibility signaling logic.

Cross-country regulatory harmonization can leverage knowledge graph comparability. As EU CSRD, ISSB standards and U.S. SEC rules converge, a shared action-level taxonomy (topics, actors, modalities) would enable global benchmarking without imposing uniform ratings. Regulators can monitor aggregate disclosure patterns to identify emerging issues(e.g. sudden increases in data security actions could signal sector-wide cyber threats) or evaluate policy efficacy (e.g. did diversity disclosure requirements shift modality from future_commitment to ongoing?).

Finally, this approach democratizes ESG analysis. Third-party ratings are costly proprietary products inaccessible to small investors, civil society organizations, or developing-country regulators. Open-sourcing knowledge graph construction methodologies and taxonomies can build public ESG infrastructure, analogous to financial accounting standards. Regulators might mandate machine-readable disclosure formats (structured ESG action reports alongside narrative documents), enabling automated knowledge graph construction and lowering stakeholder analysis costs.

5.3.3.1. Standardized disclosure template recommendation. To operationalize modality-based credibility assessment and enable automated compliance monitoring, regulators could mandate a tabular action reporting format mirroring the 8-element extraction schema (Table 5).

This structured format enables cross-firm comparability without prescribing disclosure content, balances flexibility with consistency, and facilitates automated extraction for regulatory oversight and public ESG infrastructure development.

5.4. Limitations and boundary conditions

5.4.1. Methodological scope: disclosure traceability versus action veracity

This framework ensures that extracted actions are accurately traceable to source text with high reliability (89.5% accuracy, Cohen's $\kappa = 0.95$ for inter-rater agreement). However, traceability to disclosed content does not constitute verification of real-world action occurrence or impact. We document what companies claim to do, not what they actually accomplish. This distinction is fundamental: high extraction accuracy means we correctly identify disclosed claims, but disclosure accuracy does not guarantee action veracity. The subsequent limitations detail the implications of this boundary condition.

5.4.2. Disclosure versus performance gap

A fundamental limitation of disclosure-based analysis is the potential gap between what companies disclose and what they actually do. This knowledge graph captures disclosed actions with high accuracy (89.5% validation accuracy, Cohen's $\kappa = 0.95$), but disclosure accuracy does not guarantee action veracity.

For example, a company may accurately disclose 'achieved 30% emission reduction' while cherry- picking favorable metrics (Scope 1–2 only, ignoring Scope 3), using questionable baselines (comparing to peak emissions rather than trend), or lacking third-party verification. Such strategic disclosure can- not be detected without external validation data (e.g. carbon registries, regulatory filings, third-party audits).

This limitation is inherent to self-reported disclosure analysis and affects all disclosure-based re- search (Cao et al., 2023; Christensen et al., 2022). This methodology reveals *what companies claim* in sustainability reports, not *what companies achieve* in actual operations. Stakeholders seeking performance evaluation should triangulate these disclosure-level findings with external performance databases (e.g. CDP climate data, MSCI ESG ratings for outcome verification), regulatory filings (e.g. mandatory emissions reporting where available) and third-party assurance reports.

Future research should develop integrated frameworks combining disclosure-level knowledge graphs (capturing claims and commitments) with performance-level datasets (validating outcomes and impacts). Such tri- angulation would enable systematic assessment of disclosure-performance alignment, identify persistent gaps signaling potential greenwashing, and support more robust ESG due diligence.

Data Source Scope: Formal Sustainability Reports *versus* Multi-Source Triangulation. The dataset is limited to formal sustainability reports (integrated reports, CSR reports, and dedicated ESG/sustainability disclosures) published by companies. This dataset does not incorporate additional public data sources such as regulatory filings and mandatory disclosures (e.g. securities reports, environmental permits), third-party verification reports and assurance statements, enforcement records and compliance violations, or media coverage and stakeholder complaints. This design choice reflects deliberate trade-offs between *data consistency* and *information completeness*.

Formal sustainability reports provide standardized narrative structures, enabling scalable LLM extraction with consistent schema adherence. However, they represent self-curated disclosures subject to strategic framing and selective emphasis. In contrast, regulatory filings and enforcement records contain mandatory, externally verified data that could validate or contradict voluntary disclosures, while media sources capture stakeholder perceptions and controversy exposure that companies may underreport.

Future extensions could incorporate multi-source triangulation by cross-referencing disclosed emission reduction claims with mandatory greenhouse gas registry data, enriching governance action records with proxy voting results and board composition filings, and detecting disclosure-reality gaps by comparing reported labor practices with inspection records or news coverage of workplace incidents. Such integration would require developing entity resolution and temporal alignment methods to reconcile cross-source references, along with source credibility weighting schemes to handle contradictory information. LLM Extraction: Inherent Biases, Model Sensitivity, and Error Propagation. While these validation results(89.5% accuracy, Cohen's $\kappa = 0.95$) demonstrate reliable extraction quality, analysis must acknowledge inherent limitations of LLM-based information extraction that warrant future investigation. *Potential Biases Introduced by LLMs.* LLMs may introduce systematic biases through over-standardization and culturally specific term misinterpretation. Over-standardization bias occurs when LLMs normalize culturally specific practices into generic categories – for instance, Japanese continuous improvement practices like 'kaizen' or 'genba' may be translated as generic 'improvement' or 'on-site management', losing cultural nuance and organizational specificity. Linguistic ambiguity handling poses challenges particularly for Japanese corporate disclosures, where implicit grammatical subjects require contextual inference that may not always align with company intent.

This approach partially mitigates these biases through bilingual processing (validating Japanese originals against English translations to detect semantic drift) and domain-specific prompt engineering (instructing the LLM to preserve organization-specific terminology). However, these biases have not been exhaustively quantified residual bias across all cultural dimensions. Analysis suggests that such biases primarily affect *semantic granularity*(losing cultural specificity) rather than *structural accuracy* (misidentifying action existence or topic assignment), but systematic bias audits comparing LLM extractions with human expert annotations across cultural dimensions remain a valuable direction for future research.

5.4.2.1. Sensitivity to model choice. The pipeline employs GPT-4o for its strong multilingual capabilities and structured output adherence. However, this study has not conducted comprehensive multi-model comparisons to assess how results might vary with alternative models (e.g. Claude 3.5 Sonnet, Gemini 1.5 Pro, open-source models like Llama 3.1). Theoretically, model choice could affect extraction quality through training corpus differences (models trained heavily on English corporate documents may underperform on Japanese sustainability terminology), instruction-following precision (ability to adhere strictly to the 8-element action schema), and confidence calibration (reliability of self-assessed extraction confidence scores).

Analysis suggests that *aggregate patterns* – such as topic distributions, modality breakdowns and stakeholder emphasis – are likely robust to model choice, as these reflect genuine disclosure patterns rather than model-specific artifacts. However, *granular extraction accuracy* (precise boundary detection for action descriptions, nuanced modality classification) may exhibit model sensitivity. Establishing this empirically through systematic multi-model benchmarking, ideally with expert-annotated gold-standard datasets, would strengthen methodological generalizability.

5.4.2.2. Error propagation to downstream analyses. Extraction errors – even at the relatively low error rate (10.5% based on validation sample) – can propagate through network analysis and clustering with potentially amplified effects. In network analysis, extraction errors affecting high-degree nodes (frequently mentioned topics or stakeholders) could distort centrality metrics more severely than errors in peripheral nodes. For example, if 'employees' as a stakeholder entity is systematically under-extracted in certain report sections, employee centrality in stakeholder networks would be underestimated, potentially misrepresenting organizational priorities.

In archetype clustering, extraction errors could cause archetype reassignment if errors cluster non-randomly. For instance, if future-oriented commitment statements are systematically harder to extract than ongoing actions (due to conditional language complexity), firms genuinely emphasizing forward-looking commitments might be misclassified into action-oriented archetypes. The modality analysis (Table 6) shows clear preferences (65.5% ongoing, 19.1% future), but analysis cannot rule out that some of this imbalance reflects extraction difficulty rather than genuine disclosure patterns.

This approach partially addresses propagation risks through confidence-based filtering (excluding low-confidence extractions, mean confidence = 0.89) and aggregate-level validation (these findings align with prior Japanese ESG research regarding social emphasis (Ioannou & Serafeim, 2017; Hogan et al., 2022), suggesting systematic extraction errors are limited). However, systematic robustness checks – such as re-running network analysis and clustering on high-confidence subsets (e.g. confidence > 0.95) and assessing result stability, or conducting sensitivity analyses with simulated error injection – remain beyond the current scope but warrant future investigation.

5.4.2.3. Limitations as future research agenda. These limitations emphasize that these limitations represent research opportunities rather than fatal methodological flaws. This study establishes *proof of concept* that LLM-driven knowledge graphs can extract meaningful ESG patterns at scale, provides a *replicable framework* with transparent validation, and uncovers *substantive empirical patterns* (social dominance, stakeholder structures, strategic archetypes) that align with theoretical expectations and prior research. Comprehensive bias audits, multi-model robustness benchmarking and error propagation sensitivity analyses – while valuable – are extensive research programs in themselves that extend beyond the scope of demonstrating methodological feasibility and initial empirical application.

As the ESG AI research community matures, establishing standardized evaluation protocols (expert-annotated multilingual ESG corpora, multi-model benchmarking suites, bias audit frameworks) will be essential infrastructure. The open-sourced taxonomy and extraction schema can serve as starting points for such collective efforts.

6. Conclusion

This study demonstrates that LLM-driven knowledge graphs can deconstruct corporate ESG narratives into granular, analyzable actions, offering a transparent and scalable alternative to opaque third-party ratings. By extracting 77,734 action records from 141 Japanese firms, analysis reveals that Japanese ESG disclosure is social-dominant (55.4%), action-oriented (65.5% ongoing *versus* 19.1% future commitments), and

employee-centric in stakeholder structure. Network analysis uncovers integrated ESG strategies with blurred E-S-G boundaries and identifies three strategic archetypes (balanced, social-oriented, environment-oriented).

This contribution is threefold. Theoretically, this study extends stakeholder salience and signaling theories by providing action-level empirical validation of disclosure patterns. Methodologically, this research establishes a replicable AI-driven framework that transforms unstructured sustainability narratives into structured knowledge graphs, demonstrating how artificial intelligence can enhance disclosure transparency to address ESG evaluation's credibility crisis. Importantly, this approach complements rather than replaces evaluative ESG ratings – the framework provides the verifiable, granular evidence layer that enables stakeholders to conduct context-specific assessments, while rating agencies provide aggregated performance benchmarks. Practically, this methodology equips investors with due diligence tools, firms with benchmarking capabilities, and regulators with policy evaluation evidence.

As ESG disclosure becomes mandatory across jurisdictions (EU CSRD, ISSB standards), the volume of unstructured sustainability reports will overwhelm human analytical capacity. AI-driven knowledge extraction is not merely a research tool but an infrastructural necessity for sustainability's data economy.

Three strategic directions emerge for extending this framework to build a global, dynamic, granular ESG knowledge commons.

Longitudinal Commitment Tracking. The action-level schema explicitly captures modality (ongoing, future_commitment, past_action), creating a foundation for temporal analysis to track whether disclosed commitments are fulfilled over time. By collecting multi-year sustainability reports and linking actions through entity resolution – matching forward-looking commitments (e.g. 'achieve net-zero by 2050' in 2024 report) to subsequent progress claims (e.g. 'reduced emissions by 15%' in 2025 report) – researchers can quantify commitment-delivery gaps and identify persistent greenwashing patterns where high future commitments are not followed by concrete ongoing actions. Technical challenges include developing action-matching algorithms robust to linguistic variation ('will reduce emissions' *versus* 'emission reduction achieved') and handling structural breaks when companies change reporting frameworks. This longitudinal capability would enable corporate ESG credibility scoring based on historical commitment fulfillment rates, providing investors with dynamic rather than static assessments.

Multimodal Data Integration. Sustainability reports contain rich non-textual information – emission trend charts, supply chain metrics tables, stakeholder engagement infographics – that the current text-only pipeline cannot process. Future extensions could integrate vision-language models (e.g. GPT-4 Vision, Gemini Pro Vision) to extract structured data from images and tables, or develop specialized PDF table parsers for quantitative disclosures. This would enable complete action verification by matching textual claims ('reduced water consumption by 30%') against visual evidence (water usage trend charts), detecting claim-evidence inconsistencies that signal potential misreporting. The resulting fully quantified knowledge graphs would link each action node to numerical performance metrics, confidence intervals, and source visuals, substantially enhancing disclosure verifiability. Technical challenges include handling diverse chart types, resolving unit inconsistencies across tables and establishing semantic alignment between text and visuals.

Cross-Country Comparative Analysis. The Japan-focused taxonomy reflects local regulatory contexts (Corporate Governance Code emphasis, stakeholder capitalism norms) and disclosure conventions. Extending to other jurisdictions requires culturally adaptive taxonomies that preserve core ESG dimensions while accommodating regional priorities – for instance, incorporating Modern Slavery Act disclosures for UK firms, detailed Scope 3 supply chain emissions mandated by EU CSRD, or community investment metrics emphasized in emerging markets. Methodologically, this demands multilingual LLM benchmarking to validate extraction quality across languages (assessing whether English-centric models underperform on non-English reports), cross-cultural prompt engineering to adapt semantic understanding to diverse reporting norms, and metadata-driven taxonomy alignment to map regional topics onto global ESG frameworks (GRI, SASB, TCFD). Successfully addressing these challenges would enable truly global ESG knowledge graphs supporting cross-border investment decisions, international policy harmonization and comparative sustainability research.

These extensions collectively position this framework as foundational infrastructure for a broader research agenda, transforming corporate sustainability from a realm of opaque ratings and contested narratives into a domain of transparent, comparable, actionable intelligence.

Transparent and accurate measurement is foundational to sustainable management. This study takes one step toward making corporate sustainability measurable, transparent and actionable at scale.

CRediT authorship contribution statement

Tong Xu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Visualization. Hideo Suzuki: Writing – Review & Editing, Supervision, Project administration, Validation.

Note

1. Company identity omitted per standard research practice; all data verifiable in our dataset.

Ethics statement

This research analyzes publicly available corporate sustainability reports and does not involve human subjects. All data sources are cited and accessible. No ethical approval was required.

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Data availability statement

The data that support the findings of this study are available as described below, in compliance with Taylor & Francis's data sharing policy.

Corporate Sustainability Reports (Publicly Available): All 141 sustainability reports analyzed in this study are publicly accessible via GitHub Release(v1.0-data). The dataset comprises two ZIP archives containing PDF files totaling 2.3 GB, covering Japanese listed companies for fiscal year 2024 (January–March 2024 collection period). Reports were originally downloaded from companies' official investor relations or corporate sustainability disclosure websites. A complete list of company names, report titles and publication dates is provided in Supplementary Appendix G, Table G1.

LLM Extraction Prompts and Methodology (Publicly Available): All LLM prompts, extraction protocols and methodological documentation are publicly available via GitHub at https://github.com/AAAIstong/esg-action-extraction. The repository includes: (1) Three-stage prompt templates (weak extraction, strong verification, RAG-enhanced validation) in YAML format; (2) JSON schema for eight-element action records; (3) Sample input/output examples; (4) Detailed extraction pro- tocol documentation describing RAG implementation, confidence scoring and validation procedures.

Extracted ESG Action Dataset (Upon Reasonable Request): The structured dataset of 77,734 ESG action records (comprising company, topic, actor, action, target, modality, evidence text and confidence score for each record) is available from the corresponding author upon reasonable request. Due to the dataset's size (approximately 45 MB), computational intensity of the extraction process (USD 2400 Open AI API costs), and to ensure appropriate academic

use, the dataset is not publicly deposited but can be shared for non-commercial research purposes subject to a data use agreement. Requests should specify: (1) intended research use; (2) institutional affiliation; (3) confirmation of non-commercial purpose. Response time: typically 2–4 weeks.

Knowledge Graph Construction Code (Upon Reasonable Request): Neo4j Cypher queries for knowledge graph construction, network analysis scripts (Louvain clustering, centrality calculations), and Python data processing pipelines are available from the corresponding author upon reasonable request. These materials require Neo4j database infrastructure and substantial computational resources, hence are shared selectively for academic replication studies.

Contact for Data Requests: Researchers wishing to access the extracted dataset, replication code, or methodological clarifications should contact the corresponding author (Tong Xu.) at xutongshenqing@163.com. All requests will be reviewed for compliance with non-commercial research use and institutional affiliation requirements. This commitment to promoting transparency and reproducibility in ESG disclosure research while protecting proprietary computational infrastructure.

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